



ITSIMPLERA BLUE TEAM INTERNSHIP

NAME : IQRA BATOOL

**REGISTRATION NUMBER :
SOCB01-6046**

WEEK 01

Objective

The purpose of this internship task is to learn the basics of a Security Operations Center (SOC) by creating a small virtual lab using Wazuh SIEM. The lab helps in monitoring systems, collecting logs, deploying Wazuh agents, and detecting security events. It also provides practical experience with File Integrity Monitoring (FIM) and alert verification in a simple and beginner-friendly environment.

Introduction

During this internship, a virtual cybersecurity lab was created using VMware. Three virtual machines were used: Ubuntu Server, Kali Linux, and Windows. Ubuntu Server hosts the Wazuh platform, Kali Linux is used for testing, and Windows acts as a client machine. This setup provides hands-on experience with basic SOC operations.

Virtual Machine Installation

Kali Linux Installation

Introduction

Kali Linux is a Linux operating system specially designed for cybersecurity and ethical hacking. It includes many built-in security tools used for penetration testing and vulnerability assessment. In this project, Kali Linux is used as the testing machine to generate security events. It helps create a safe environment for learning and practicing cybersecurity skills.



Purpose of Installation

The main purpose of installing Kali Linux is to perform security testing in the virtual lab. It acts as the attacker machine and allows testing without affecting the host computer. For better performance, the virtual machine was assigned **2 CPU cores, 2 GB RAM, and 40 GB storage**. These resources are enough to run Kali Linux and its tools smoothly.

Installation Process

The official Kali Linux ISO file was downloaded from the official website. A new virtual machine was created in VMware, and the ISO file was selected for installation. The required CPU, RAM, and storage were allocated during the setup process. Two network adapters, **NAT** and **Host-Only**, were also configured to provide internet access and communication with other virtual machines.

Verification

After completing the installation, the network settings were checked to make sure both adapters were working correctly. IP addresses were verified, and the **ping** command was used to test communication with other virtual machines and the internet. This confirmed that the Kali Linux machine was ready for security testing and future Wazuh monitoring tasks.

Ubuntu Server Installation

Introduction

Ubuntu Server is a lightweight and reliable Linux operating system used for running servers and network services. In this project, it is used to install the Wazuh platform for security monitoring. It works as the main server that collects and analyzes logs from other virtual machines. This makes it an important part of the SOC lab.



Purpose of Installation

The Ubuntu Server virtual machine is installed to act as the central server in the lab. It hosts the Wazuh platform and receives logs from connected systems. It also analyzes security events and helps monitor network activity. This setup provides a practical environment for learning SOC operations.

Resource Allocation

Ubuntu Server was configured with **2 CPU cores, 8 GB RAM, and 40 GB storage** to ensure smooth performance. The official Ubuntu Server ISO was downloaded and installed on VMware. During the installation, NAT and Host-Only network adapters were added to provide internet access and communication with other virtual machines. A username, password, and other basic settings were also configured.

Verification

After the installation was completed, the network configuration was verified by checking the assigned IP addresses. The **ping** command was used to test communication with the internet and the other virtual machines. All tests were successful, confirming that the server was working properly. Ubuntu Server is now ready to run Wazuh and monitor security events in the virtual lab.

Windows 10 Installation

Introduction

Windows 10 is one of the most popular operating systems used in homes, offices, and organizations. In this virtual lab, it is used as a client machine to generate normal user activities. It also helps test the Wazuh monitoring system in a real-world environment. This makes the lab setup more practical and realistic.



Purpose of Installation

The main purpose of installing Windows 10 is to represent a normal user computer in the SOC lab. It allows the Wazuh Agent to monitor system activities and collect security logs. The virtual machine was assigned **2 CPU cores, 4 GB RAM, and 50 GB storage**, which is enough for smooth performance. This setup helps in understanding how Windows endpoints are monitored in an organization.

Installation Process

The official Windows 10 ISO file was downloaded from Microsoft's website and installed using VMware Workstation. During the setup, the required CPU, RAM, and storage were allocated to the virtual machine. Two network adapters, **NAT** and **Host-Only**, were configured to provide internet access and communication with other virtual machines. After the installation, a user account and basic system settings were completed.

Verification

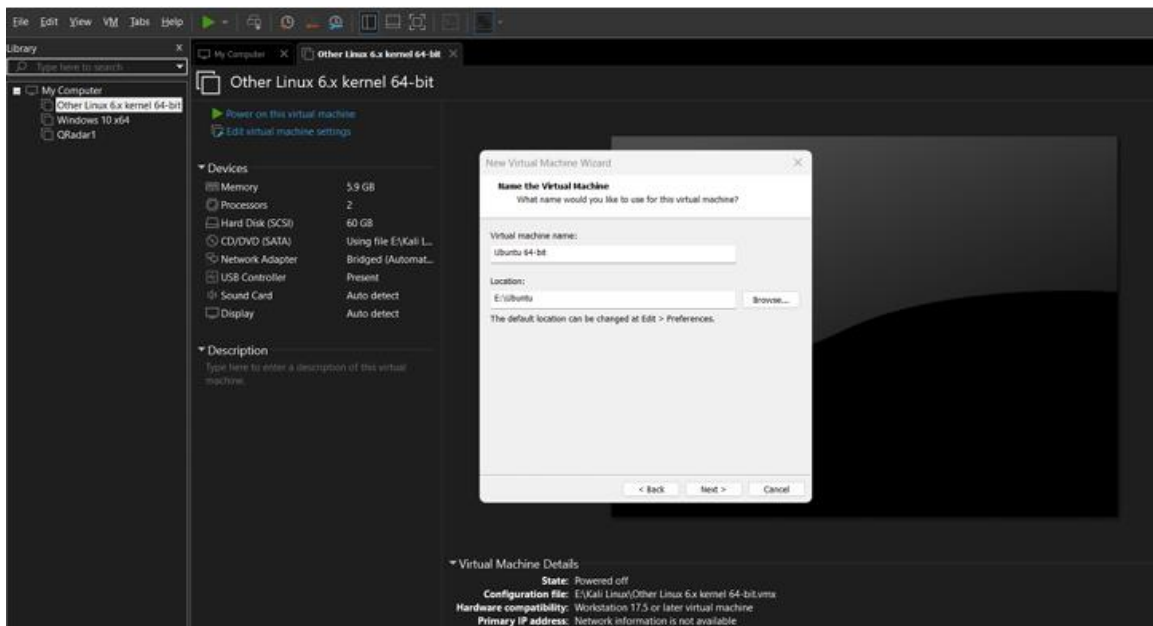
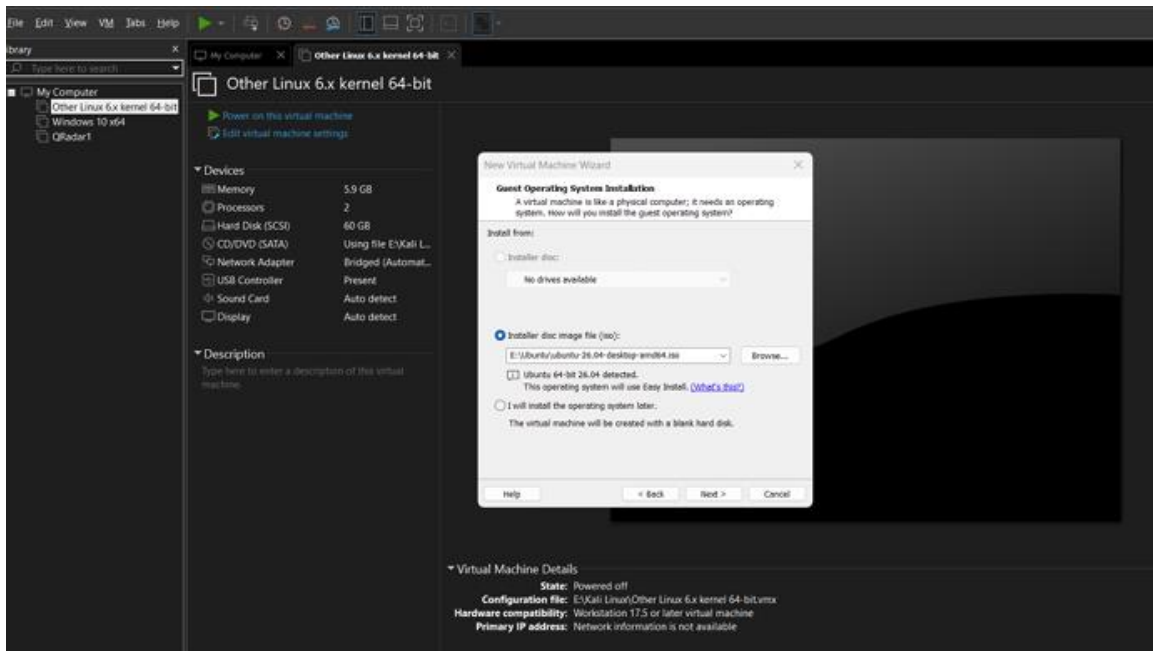
Once the installation was finished, the network adapters were checked to ensure they received valid IP addresses. The connection with other virtual machines and the internet was tested using the **ping** command. All tests were successful, confirming that the Windows virtual machine was working correctly. It is now ready for Wazuh Agent installation and security monitoring.

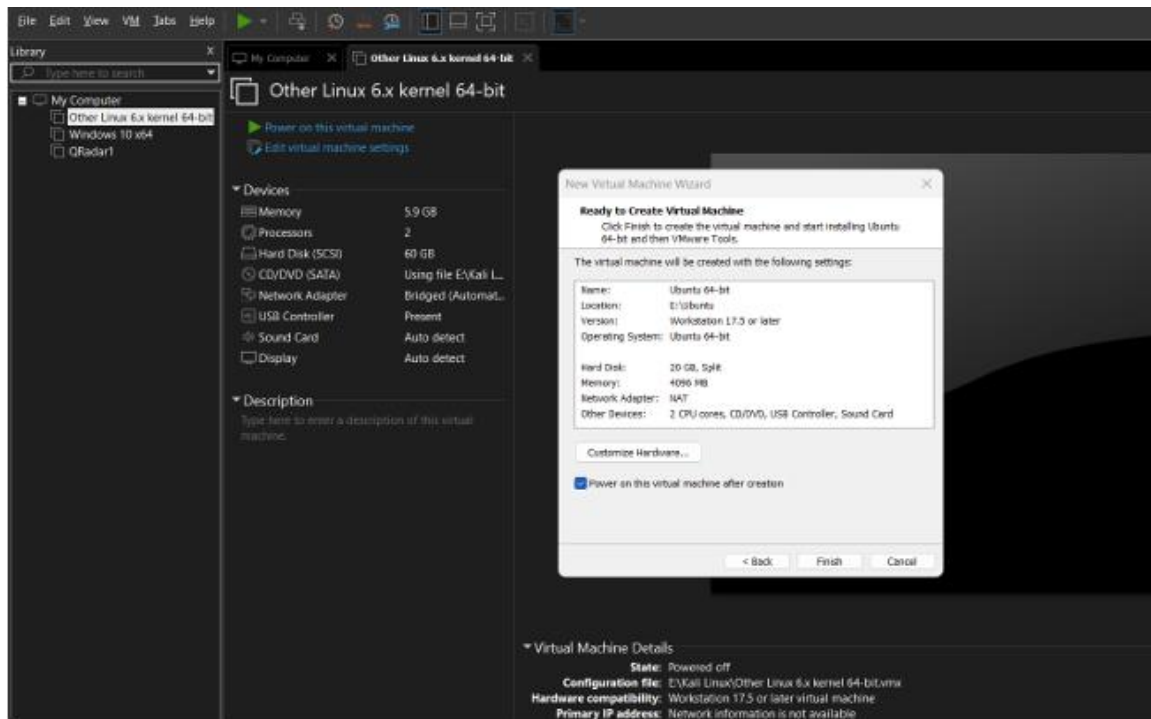
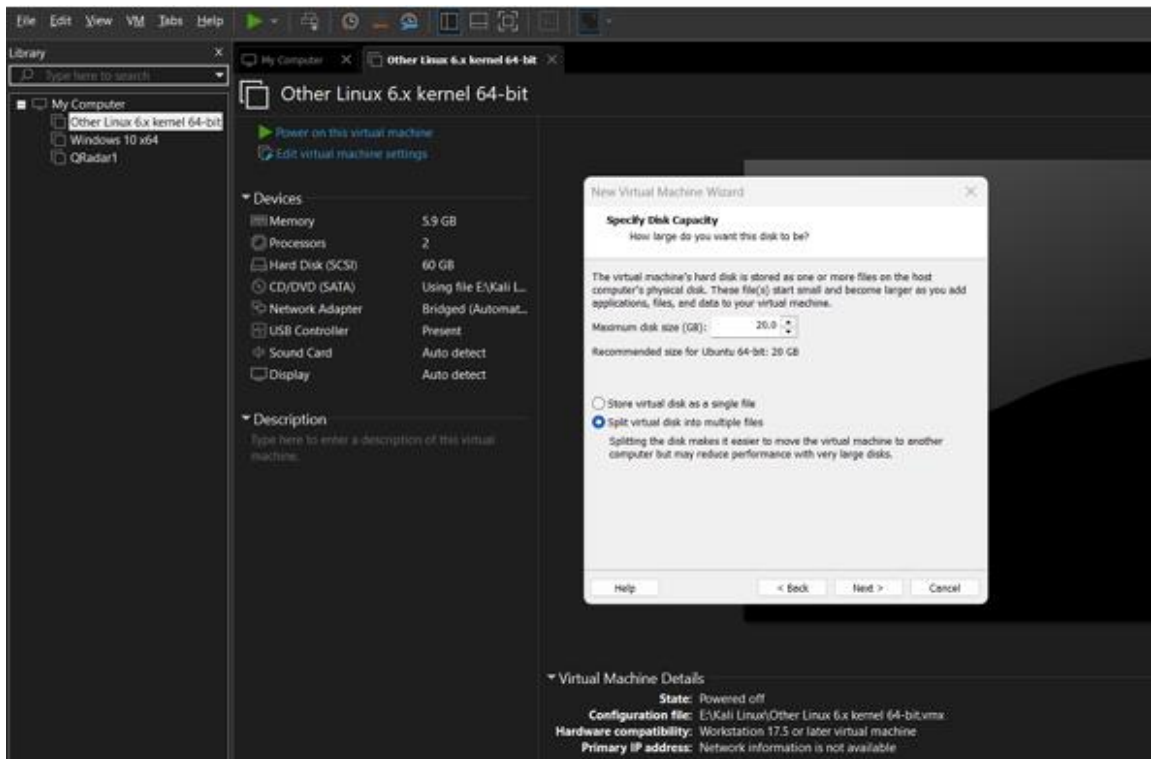
Practical Part

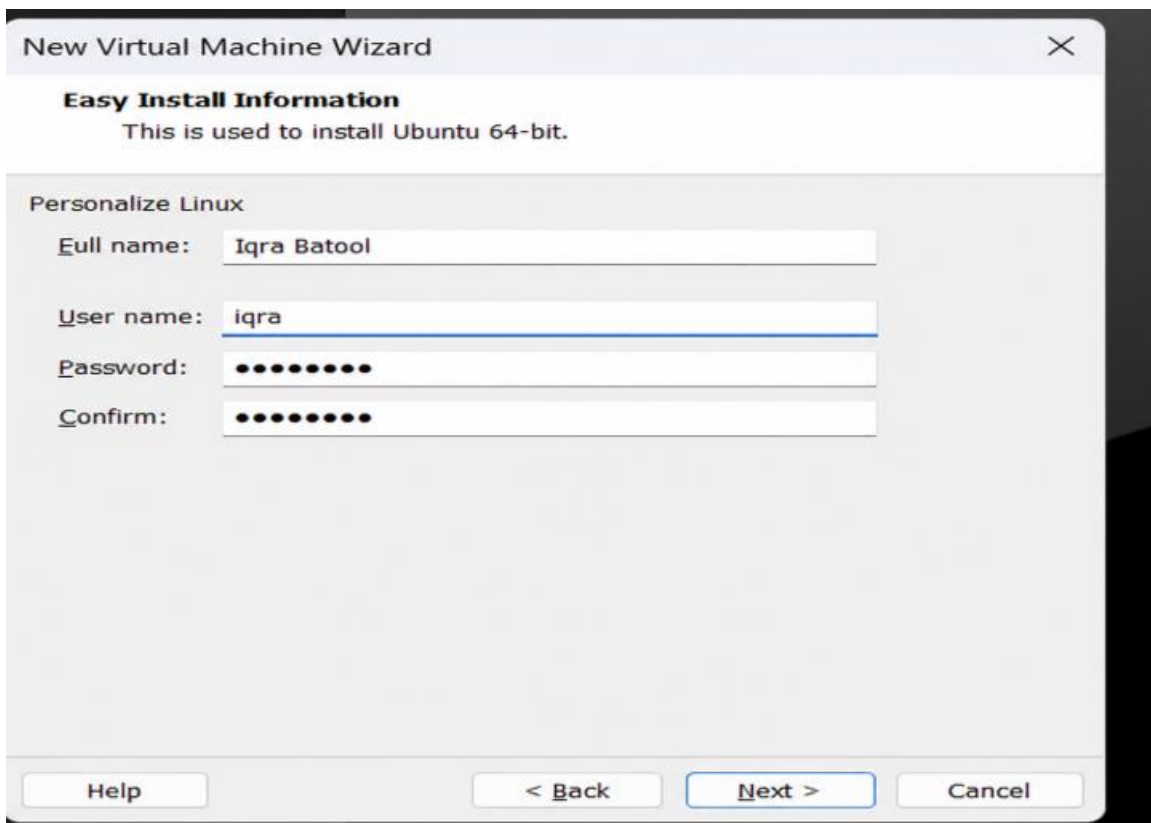
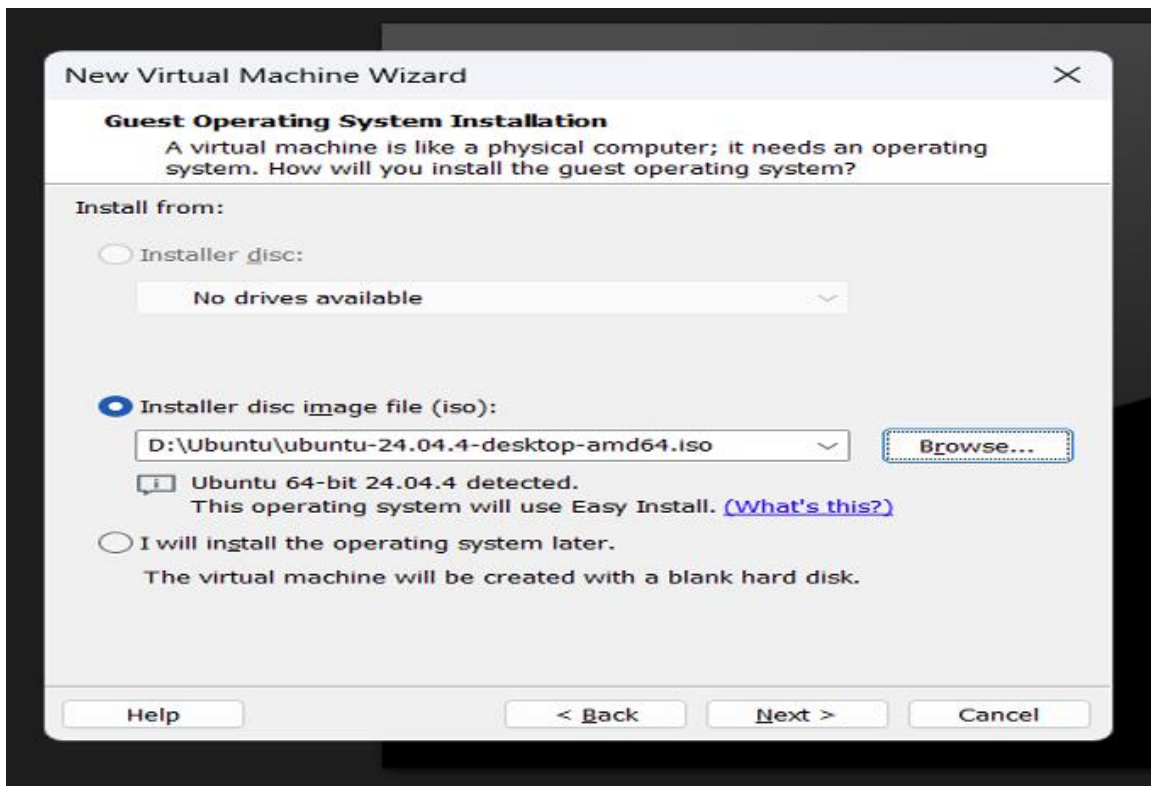
To create the virtual lab, **Ubuntu 24.04.4 Desktop (64-bit)** was downloaded from the official Ubuntu website. The ISO file was then used to install Ubuntu on **VMware Workstation** as a virtual machine. After the installation, the basic system settings and network configuration were completed successfully. This virtual machine was prepared for Wazuh installation and other cybersecurity lab activities.

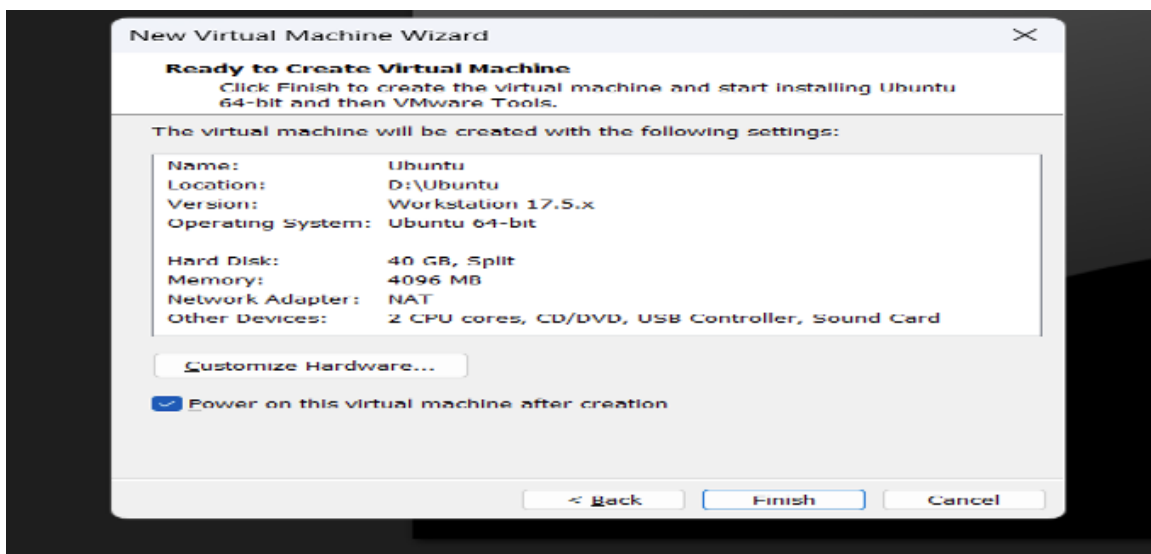
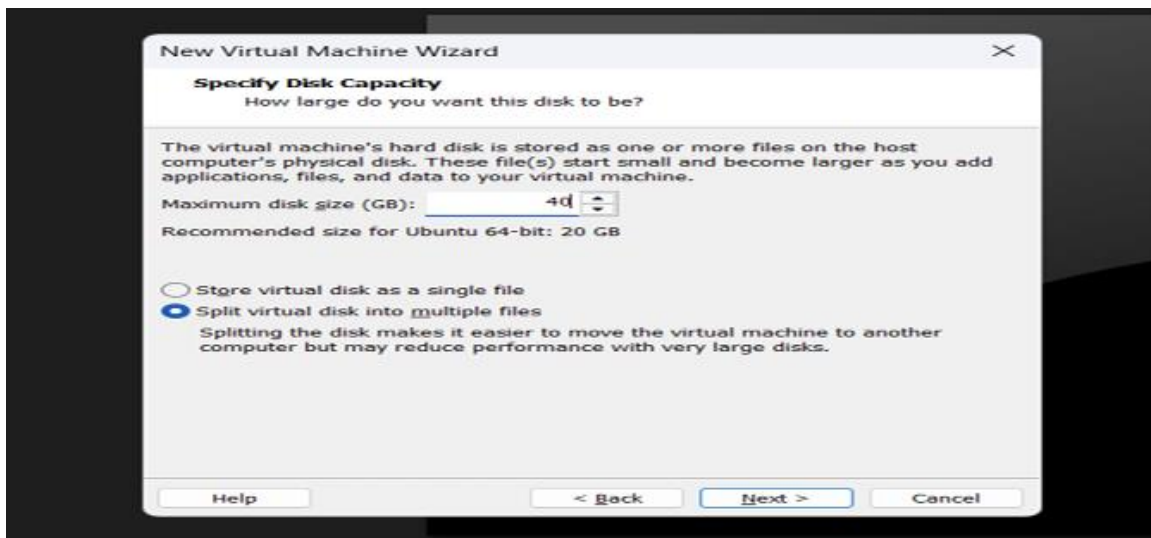
Installation Steps

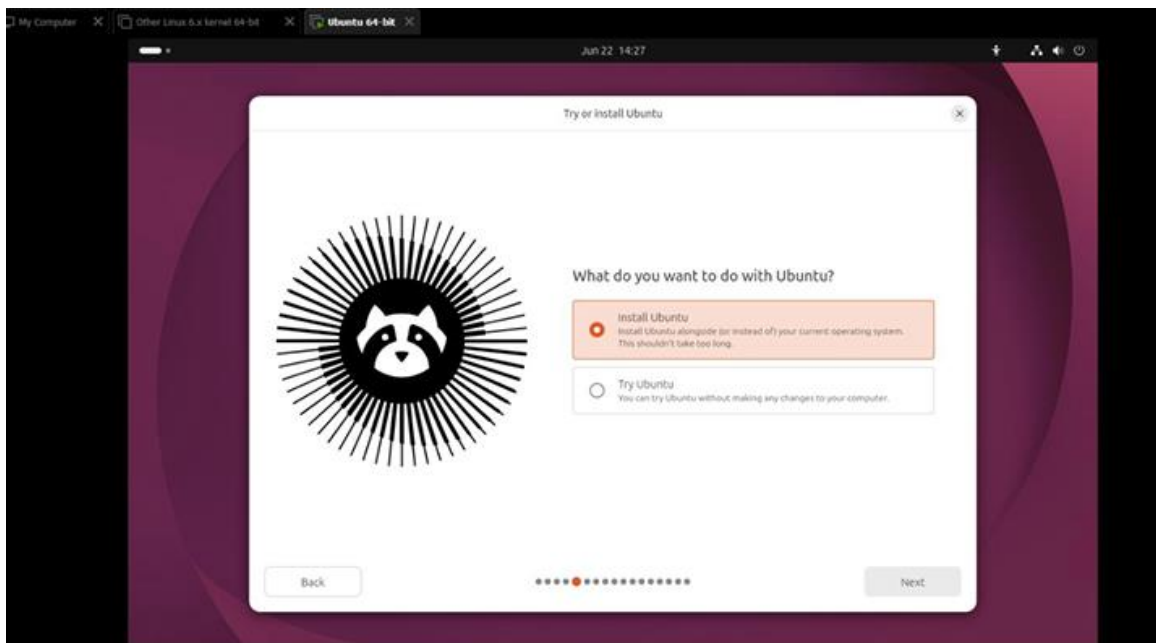
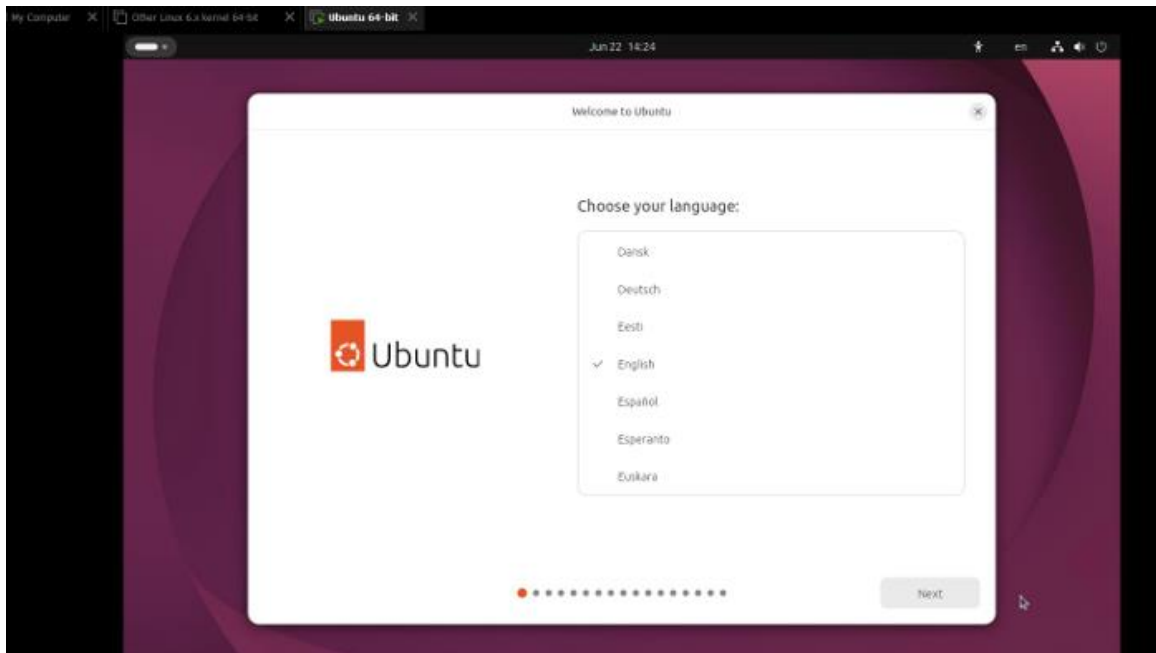
- Open **VMware Workstation** and click **Create a New Virtual Machine**.
- Browse and select the downloaded **Ubuntu 24.04.4 Desktop ISO** file.
- Continue the installation by following the setup wizard.
- Configure the required system settings and complete the installation.
- The following screenshots show each step of the installation process.











Jun 29 13:08

Create your account



Create your account

Your name
Iqra Batool ✓

Your computer's name
iqra-batool-VMware-Virtual-Platform ✓

Your username
iqra-batool ✓

Password
•••••• Show Fair password

Confirm password
•••••• ✓

☒ Require my password to log in

☐ Use Active Directory

Back Next

File Edit View VM Tools Help

Library

- My Computer
 - Other Linux 6.x kernel 64-bit
 - Windows 10 x64
 - QRadar1
 - Ubuntu 64-bit

Power on this virtual machine

Edit virtual machine settings

Virtual Machine Settings

Hardware Options

Device	Summary
Memory	4 GB
Processors	2
Hard Disk (SCSI)	20 GB
CD/DVD 2 (SATA)	Using file E:\Ubuntu\ubuntu...
CD/DVD (SATA)	Using file autostart.iso
Floppy	Using file autostart.flp
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Display	Auto detect

Memory

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

Memory for this virtual machine: 4096 MB

128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB

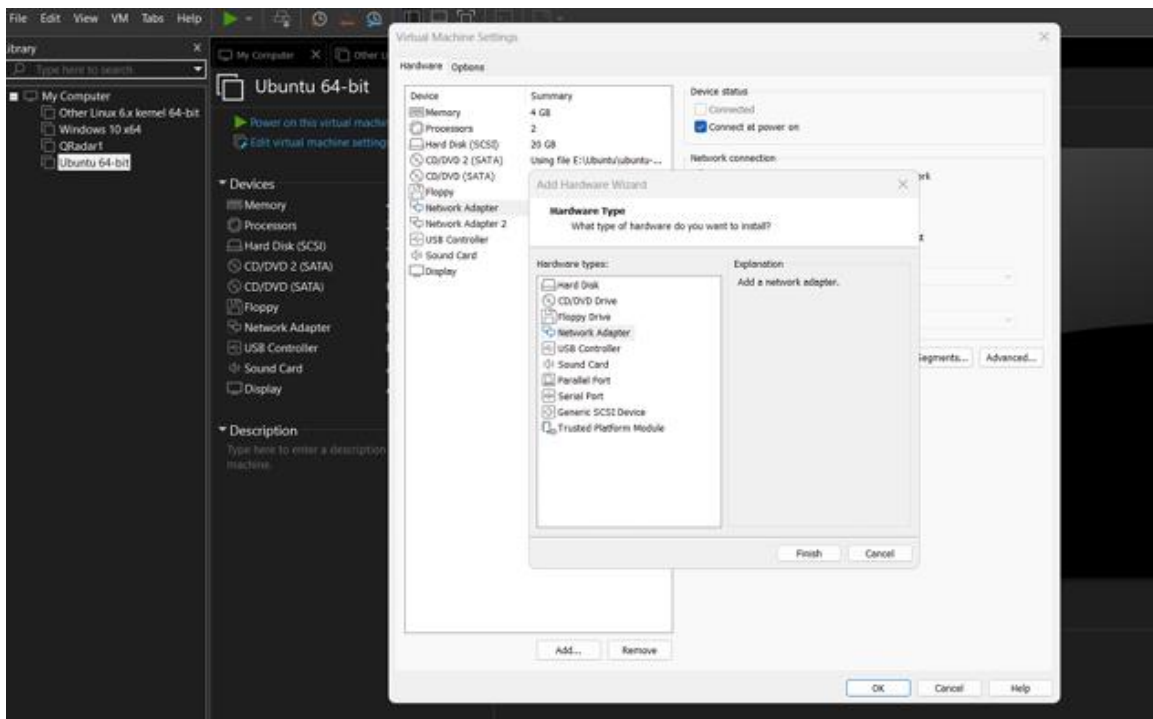
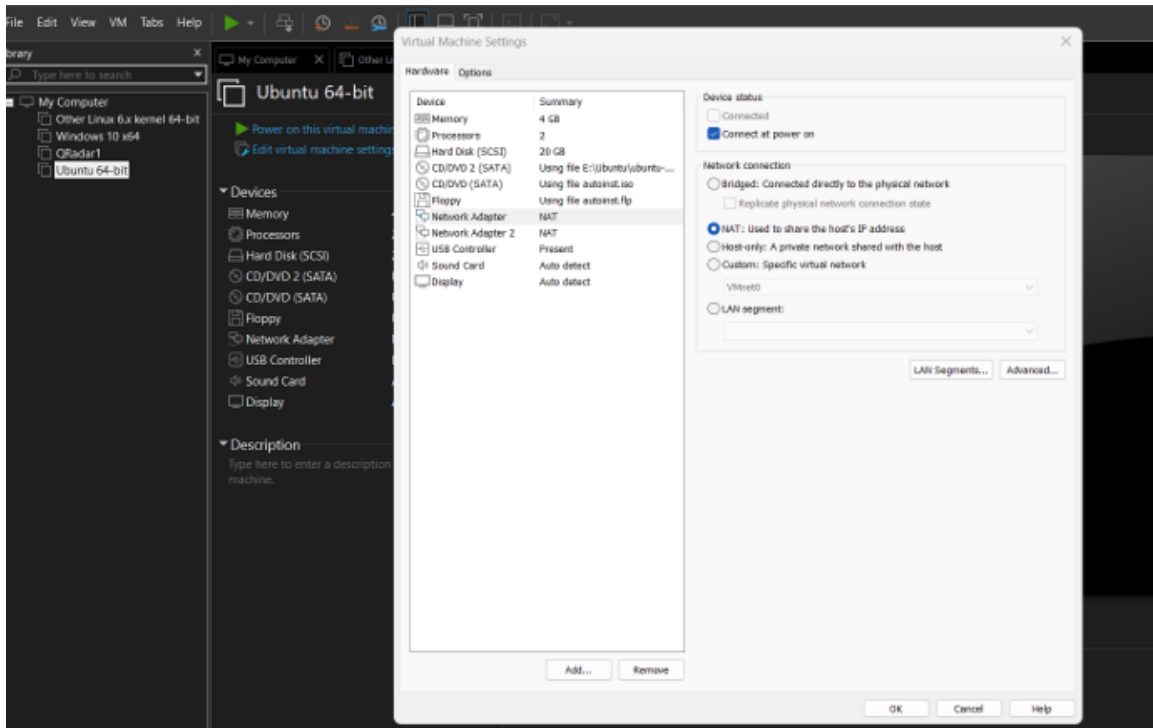
Maximum recommended memory
(Memory swapping may occur beyond this size.)
12.9 GB

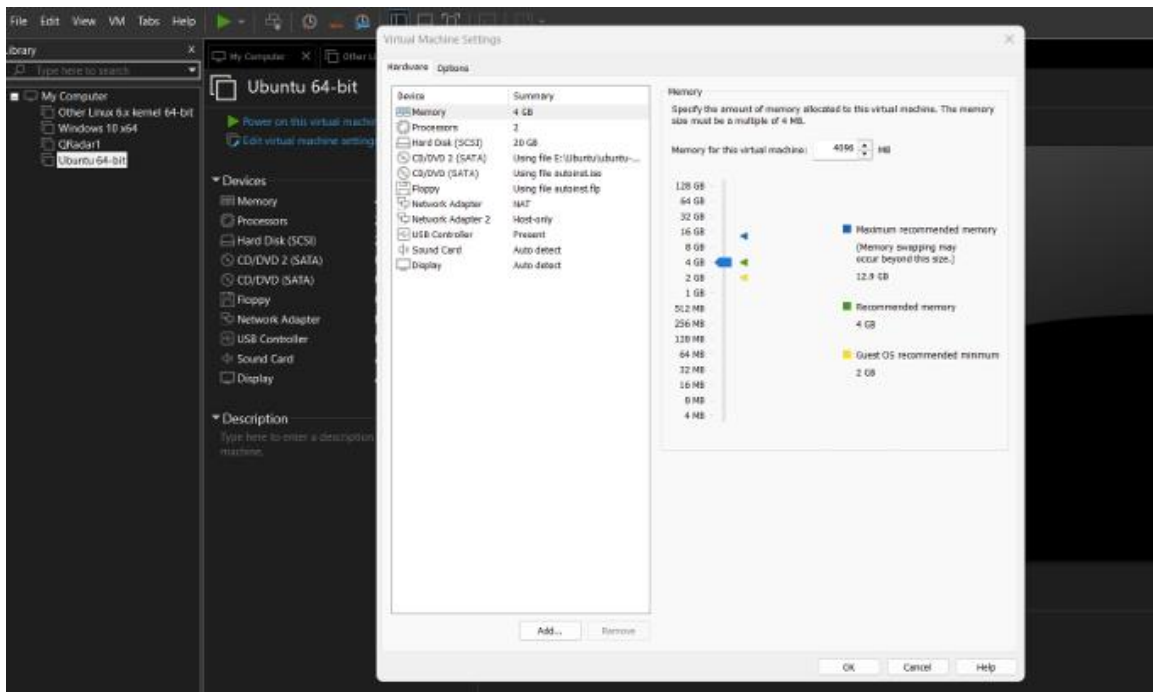
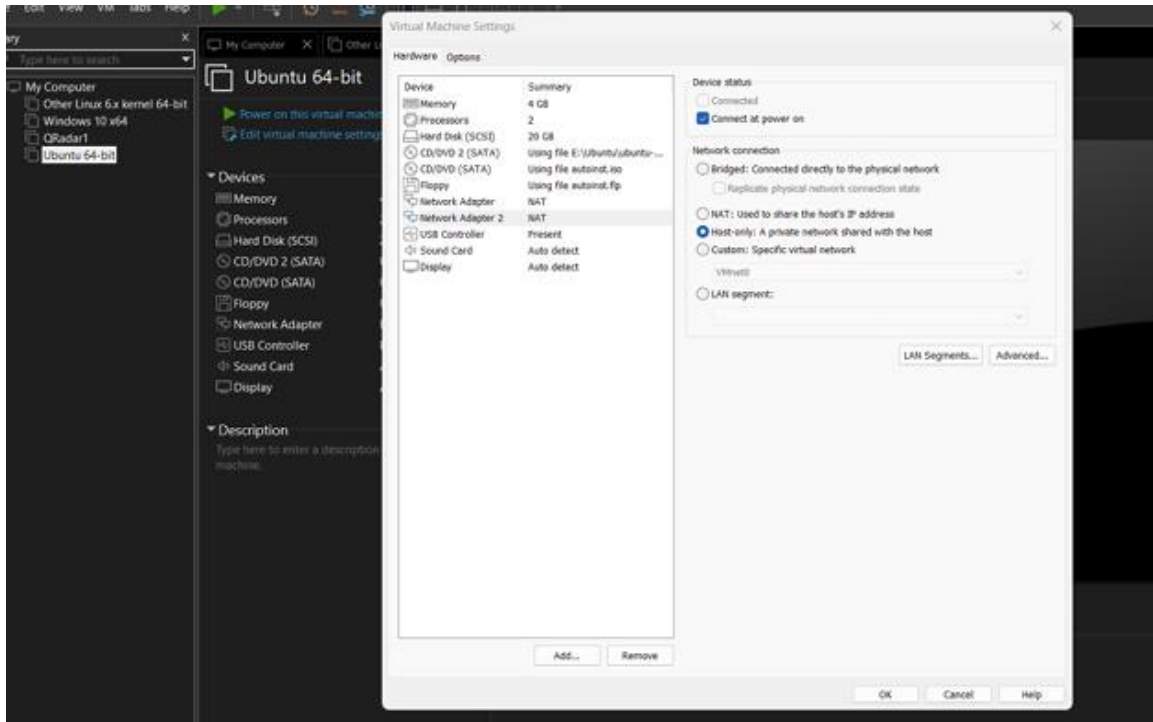
Recommended memory
4 GB

Guest OS recommended minimum
2 GB

Add... Remove

OK Cancel Help





Wazuh Installation on Ubuntu

Introduction

Security Information and Event Management (SIEM) is a security solution used to collect and monitor logs from different devices in a network. It helps detect suspicious activities and security threats by analyzing system logs. In this project, **Wazuh** is used as the SIEM platform to monitor the virtual lab.

Overview of Wazuh

Wazuh is a free and open-source security monitoring tool used in SOC environments. It helps detect threats, monitor system activity, and analyze security events. It is widely used because it is easy to install and provides useful security features for beginners and professionals.

Main Features of Wazuh

- Log collection and monitoring
- Intrusion detection
- File Integrity Monitoring (FIM)
- Vulnerability detection
- Security configuration monitoring
- Incident response

Wazuh Architecture

Wazuh works with three main components that perform different tasks together.

1. Wazuh Manager

The **Wazuh Manager** is the main component of the platform. It receives logs from connected agents, analyzes them using security rules, and generates alerts whenever suspicious activity is detected. It acts as the central monitoring system of the lab.

2. Wazuh Indexer

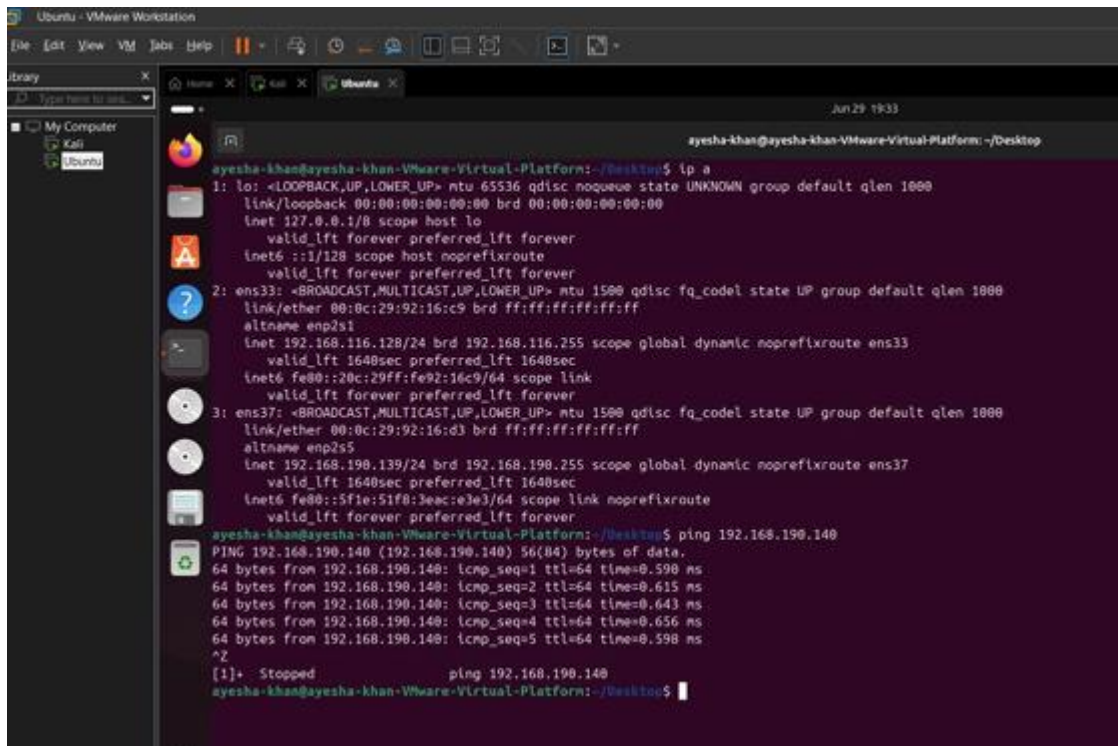
The **Wazuh Indexer** stores all logs and security alerts received from the manager. It organizes the data so it can be searched quickly whenever needed. This helps security analysts review previous events and investigate security incidents.

3. Wazuh Dashboard

The **Wazuh Dashboard** is the web interface used to manage the Wazuh platform. It displays security alerts, connected agents, system status, and collected logs in an easy-to-read format. It allows users to monitor the entire environment from one place.

Static IP Configuration

Before installing Wazuh, a **static IP address** was assigned to the Ubuntu Server. A static IP keeps the server address the same, allowing all Wazuh agents to communicate with it without connection problems. After the configuration, the installation steps were completed successful.



```
Ubuntu - VMware Workstation
File Edit View VM Jobs Help
Library
Type text to search
My Computer
Kali
Ubuntu
Jun 29 19:33
ayesha-khan@ayesha-khan-VMware-Virtual-Platform: ~/Desktop
ayesha-khan@ayesha-khan-VMware-Virtual-Platform:~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:92:16:c9 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.116.128/24 brd 192.168.116.255 scope global dynamic noprefixroute ens33
        valid_lft 1640sec preferred_lft 1640sec
    inet6 fe80::20c:29ff:fe92:16c9/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:92:16:d3 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 192.168.190.139/24 brd 192.168.190.255 scope global dynamic noprefixroute ens37
        valid_lft 1640sec preferred_lft 1640sec
    inet6 fe80::5f1e:51f8:3eac:e3e3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ayesha-khan@ayesha-khan-VMware-Virtual-Platform:~/Desktop$ ping 192.168.190.140
PING 192.168.190.140 (192.168.190.140) 56(84) bytes of data:
 64 bytes from 192.168.190.140: icmp_seq=1 ttl=64 time=0.590 ms
 64 bytes from 192.168.190.140: icmp_seq=2 ttl=64 time=0.615 ms
 64 bytes from 192.168.190.140: icmp_seq=3 ttl=64 time=0.643 ms
 64 bytes from 192.168.190.140: icmp_seq=4 ttl=64 time=0.656 ms
 64 bytes from 192.168.190.140: icmp_seq=5 ttl=64 time=0.598 ms
^Z
[1]+  Stopped                  ping 192.168.190.140
ayesha-khan@ayesha-khan-VMware-Virtual-Platform:~/Desktop$
```

WAZUH INSTALLATION PROCESS:

- For a single-node SOC lab environment, the recommended installation method is the Wazuh QuickStart installation script provided in the official documentation.

```

ayesha-khan@ayesha-khan-Virtual-Platform: ~/Desktop
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21.2 (+libidn2/2.3.7) libssh/0.10.6/openssh/
lib nghttp2/1.59.0 librtmp/2.3 OpenLDAP/2.6.10
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.9
Protocols: dict file ftp ftps gopher htp https inap inaps ldap ldaps mqtt pop3 pop3s rtsp scp sftp smb smbs smtp smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
ayesha-khan@ayesha-khan-Virtual-Platform:~/Desktop$ curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
ayesha-khan@ayesha-khan-Virtual-Platform:~/Desktop$ ls
wazuh-install.sh
ayesha-khan@ayesha-khan-Virtual-Platform:~/Desktop$ chmod +x wazuh-install.sh
ayesha-khan@ayesha-khan-Virtual-Platform:~/Desktop$ sudo ./wazuh-install.sh -o
29/06/2026 20:07:39 INFO: Starting Wazuh installation assistant. Wazuh version: 4.14.5
29/06/2026 20:07:39 INFO: Verbose logging redirected to /var/log/wazuh-install.log
29/06/2026 20:07:52 INFO: --- Dependencies ---
29/06/2026 20:08:07 INFO: Installing gawk.
29/06/2026 20:08:08 INFO: Verifying that your system meets the recommended minimum hardware requirements.
29/06/2026 20:08:08 INFO: Wazuh web interface port will be 443.
29/06/2026 20:08:23 INFO: --- Dependencies ---
29/06/2026 20:08:23 INFO: Installing apt-transport-https.
29/06/2026 20:08:30 INFO: Installing debhelper.
29/06/2026 20:09:22 INFO: Wazuh repository added.
29/06/2026 20:09:22 INFO: --- Configuration files ---
29/06/2026 20:09:22 INFO: Generating configuration files.
29/06/2026 20:09:23 INFO: Generating the root certificate.
29/06/2026 20:09:25 INFO: Generating Admin certificates.
29/06/2026 20:09:25 INFO: Generating Wazuh indexer certificates.
29/06/2026 20:09:26 INFO: Generating Filebeat certificates.
29/06/2026 20:09:28 INFO: Generating Wazuh dashboard certificates.
29/06/2026 20:09:30 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
29/06/2026 20:09:31 INFO: --- Wazuh Indexer ---
29/06/2026 20:09:31 INFO: Starting Wazuh Indexer installation.
29/06/2026 20:11:20 INFO: Wazuh indexer installation finished.
29/06/2026 20:11:20 INFO: Wazuh indexer post-install configuration finished.
29/06/2026 20:11:20 INFO: Starting service wazuh-indexer.
29/06/2026 20:12:13 INFO: wazuh-indexer service started.
29/06/2026 20:12:13 INFO: Initializing Wazuh indexer cluster security settings.

```

```

ayesha-khan@ayesha-khan-Virtual-Platform: ~/Desktop
libfwupd2
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
curl
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 227 kB of archives.
After this operation, 535 kB of additional disk space will be used.
Get:1 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.9 [227 kB]
Fetched 227 kB in 1s (205 kB/s)
Selecting previously unselected package curl.
(Reading database ... 153164 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.9_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.9) ...
Setting up curl (8.5.0-2ubuntu10.9) ...
Processing triggers for man-db (2.12.0-4build2) ...
ayesha-khan@ayesha-khan-Virtual-Platform:~/Desktop$ curl --version
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21.2 (+libidn2/2.3.7) libssh/0.10.6/openssh/
lib nghttp2/1.59.0 librtmp/2.3 OpenLDAP/2.6.10
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.9
Protocols: dict file ftp ftps gopher htp https inap inaps ldap ldaps mqtt pop3 pop3s rtsp scp sftp smb smbs smtp smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
ayesha-khan@ayesha-khan-Virtual-Platform:~/Desktop$ curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
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```



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29/06/2026 20:09:31 INFO: Starting Wazuh Indexer installation.
29/06/2026 20:11:20 INFO: Wazuh Indexer installation finished.
29/06/2026 20:11:20 INFO: Wazuh indexer post-install configuration finished.
29/06/2026 20:11:20 INFO: Starting service wazuh-indexer.
29/06/2026 20:12:13 INFO: wazuh-indexer service started.
29/06/2026 20:12:13 INFO: Initializing Wazuh Indexer cluster security settings.
29/06/2026 20:12:26 INFO: Wazuh Indexer cluster security configuration initialized.
29/06/2026 20:12:26 INFO: Wazuh Indexer cluster initialized.
29/06/2026 20:12:26 INFO: --- Wazuh server ---
29/06/2026 20:12:26 INFO: Starting the Wazuh manager installation.
29/06/2026 20:19:05 INFO: Wazuh manager installation finished.
29/06/2026 20:19:05 INFO: Wazuh manager vulnerability detection configuration finished.
29/06/2026 20:19:05 INFO: Starting service wazuh-manager.
29/06/2026 20:19:41 INFO: wazuh-manager service started.
29/06/2026 20:19:41 INFO: Starting Filebeat installation.
29/06/2026 20:20:13 INFO: Filebeat installation finished.
29/06/2026 20:20:18 INFO: Filebeat post-install configuration finished.
29/06/2026 20:20:18 INFO: Starting service filebeat.
29/06/2026 20:20:27 INFO: filebeat service started.
29/06/2026 20:20:27 INFO: --- Wazuh dashboard ---
29/06/2026 20:20:27 INFO: Starting Wazuh dashboard installation.
```

```
29/06/2026 20:09:22 INFO: --- Configuration files ---
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29/06/2026 20:20:27 INFO: filebeat service started.
29/06/2026 20:20:27 INFO: --- Wazuh dashboard ---
29/06/2026 20:20:27 INFO: Starting Wazuh dashboard installation.
29/06/2026 20:24:09 INFO: Wazuh dashboard installation finished.
29/06/2026 20:24:10 INFO: Wazuh dashboard post-install configuration finished.
29/06/2026 20:24:10 INFO: Starting service wazuh-dashboard.
29/06/2026 20:24:16 INFO: wazuh-dashboard service started.
29/06/2026 20:24:24 INFO: Updating the internal users.
```

The script automatically download , install and configures all three components of the:

- Wazuh manager
- Wazuh indexer
- Wazuh dashboard

```
29/06/2026 20:12:13 INFO: Initializing Wazuh indexer cluster security settings.
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29/06/2026 20:19:41 INFO: wazuh-manager service started.
29/06/2026 20:19:41 INFO: Starting Filebeat installation.
29/06/2026 20:20:13 INFO: Filebeat installation finished.
29/06/2026 20:20:18 INFO: Filebeat post-install configuration finished.
29/06/2026 20:20:18 INFO: Starting service filebeat.
29/06/2026 20:20:27 INFO: filebeat service started.
29/06/2026 20:20:27 INFO: --- Wazuh dashboard ---
29/06/2026 20:20:27 INFO: Starting Wazuh dashboard installation.
29/06/2026 20:24:09 INFO: Wazuh dashboard installation finished.
29/06/2026 20:24:10 INFO: Wazuh dashboard post-install configuration finished.
29/06/2026 20:24:10 INFO: Starting service wazuh-dashboard.
29/06/2026 20:24:16 INFO: wazuh-dashboard service started.
29/06/2026 20:24:24 INFO: Updating the internal users.
29/06/2026 20:24:52 INFO: A backup of the internal users has been saved in the /etc/wazuh-indexer/internalusers-backup folder.
29/06/2026 20:25:58 INFO: The filebeat.yml file has been updated to use the Filebeat Keystore username and password.
29/06/2026 20:27:20 INFO: Initializing Wazuh dashboard web application.
29/06/2026 20:27:30 INFO: Wazuh dashboard web application not yet initialized. Waiting...
29/06/2026 20:27:35 INFO: Wazuh dashboard web application not yet initialized. Waiting...
29/06/2026 20:27:52 INFO: Wazuh dashboard web application not yet initialized. Waiting...
29/06/2026 20:28:07 INFO: Wazuh dashboard web application initialized.
29/06/2026 20:28:07 INFO: --- Summary ---
29/06/2026 20:28:07 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: ?jqU6X?L8MJGHhex2gPEhA14y-8UKd7d
29/06/2026 20:28:07 INFO: --- Dependencies ---
29/06/2026 20:28:07 INFO: Removing gawk.
29/06/2026 20:28:23 INFO: Installation finished.
ayesha-khan@ayesha-khan-Virtual-Platform: ~/Desktop$
```

```
valid_lft forever preferred_lft forever
ayesha-khan@ayesha-khan-Virtual-Platform: ~/Desktop$ sudo systemctl status wazuh-manager
[sudo] password for ayesha-khan:
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-29 20:26:29 PKT; 7min ago
     Tasks: 211 (limit: 4541)
    Memory: 135.7M (peak: 832.5M swap: 295.8M swap peak: 297.8M)
       CPU: 3min 33.76ms
   CGroup: /system.slice/wazuh-manager.service
           └─61733 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             └─61734 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
               └─61735 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                 └─61738 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                   └─61741 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                     └─61783 /var/ossec/bin/wazuh-authd
                       └─61797 /var/ossec/bin/wazuh-db
                         └─61833 /var/ossec/bin/wazuh-execd
                           └─61846 /var/ossec/bin/wazuh-analysisd
                             └─61855 /var/ossec/bin/wazuh-syscheckd
                               └─61871 /var/ossec/bin/wazuh-remoted
                                 └─61908 /var/ossec/bin/wazuh-logcollector
                                   └─61927 /var/ossec/bin/wazuh-monitord
                                     └─61937 /var/ossec/bin/wazuh-modulesd

Jun 29 20:26:22 ayesha-khan-Virtual-Platform env[61668]: Started wazuh-syscheckd...
Jun 29 20:26:24 ayesha-khan-Virtual-Platform env[61668]: Started wazuh-remoted...
Jun 29 20:26:25 ayesha-khan-Virtual-Platform env[61668]: Started wazuh-logcollector...
Jun 29 20:26:25 ayesha-khan-Virtual-Platform env[61668]: Started wazuh-monitord...
Jun 29 20:26:25 ayesha-khan-Virtual-Platform env[61934]: 2026/06/29 20:26:25 wazuh-modulesd:router: INFO: Loaded router module.
Jun 29 20:26:25 ayesha-khan-Virtual-Platform env[61934]: 2026/06/29 20:26:25 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Jun 29 20:26:25 ayesha-khan-Virtual-Platform env[61934]: 2026/06/29 20:26:25 wazuh-modulesd:inventory-harvester: INFO: Loaded inventory harvester module.
Jun 29 20:26:26 ayesha-khan-Virtual-Platform env[61668]: Started wazuh-modulesd...
Jun 29 20:26:29 ayesha-khan-Virtual-Platform env[61668]: Completed.
Jun 29 20:26:29 ayesha-khan-Virtual-Platform systemd[1]: Started wazuh-manager.service - Wazuh manager.
ayesha-khan@ayesha-khan-Virtual-Platform: ~/Desktop$
```



```
Jun 29 20:26:22 ayesha-khan-Virtual-Platform env[61668]: Started wazuh-syscheckd...
Jun 29 20:26:24 ayesha-khan-Virtual-Platform env[61668]: Started wazuh-remoted...
Jun 29 20:26:25 ayesha-khan-Virtual-Platform env[61668]: Started wazuh-logcollector...
Jun 29 20:26:25 ayesha-khan-Virtual-Platform env[61668]: Started wazuh-monitord...
Jun 29 20:26:25 ayesha-khan-Virtual-Platform env[61934]: 2026/06/29 20:26:25 wazuh-modules:router: [INFO: Loaded router module.
Jun 29 20:26:25 ayesha-khan-Virtual-Platform env[61934]: 2026/06/29 20:26:25 wazuh-modules:content_manager: INFO: Loaded content_manager module.
Jun 29 20:26:25 ayesha-khan-Virtual-Platform env[61934]: 2026/06/29 20:26:25 wazuh-modules:inventory-harvester: INFO: Loaded inventory harvester module.
Jun 29 20:26:26 ayesha-khan-Virtual-Platform env[61668]: Started wazuh-modulesd...
Jun 29 20:26:26 ayesha-khan-Virtual-Platform env[61668]: Completed.
Jun 29 20:26:29 ayesha-khan-Virtual-Platform systemd[1]: Started wazuh-manager.service - Wazuh manager.
ayesha-khan@ayesha-khan-Virtual-Platform:~/Desktop$ sudo systemctl status wazuh-indexer
● wazuh-indexer.service - wazuh-indexer
   Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-29 20:12:13 PKT; 23min ago
     Docs: https://documentation.wazuh.com
    Main PID: 7597 (java)
      Tasks: 70 (limit: 4541)
     Memory: 734.9M (peak: 1.4G swap: 825.0M swap peak: 898.1M)
        CPU: 3min 39.942s
    CGroup: /system.slice/wazuh-indexer.service
           └─7597 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddress.cache.ttl=60 -Dopensearch.networkaddress.cache.negative.ttl=10

Jun 29 20:11:33 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstrap.Open
Jun 29 20:11:33 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap
Jun 29 20:11:33 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: System::setSecurityManager will be removed in a future release
Jun 29 20:11:36 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: Jun 29, 2026 8:11:36 PM sun.util.locale.provider.LocaleProviderAdapter <clinit>
Jun 29 20:11:36 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: COMPAT locale provider will be removed in a future release
Jun 29 20:11:37 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: A terminally deprecated method in java.lang.System has been called
Jun 29 20:11:37 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstrap.Secur
Jun 29 20:11:37 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap
Jun 29 20:11:37 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: System::setSecurityManager will be removed in a future release
Jun 29 20:12:13 ayesha-khan-Virtual-Platform systemd[1]: Started wazuh-indexer.service - wazuh-indexer.
(lines 1-21/21 (END))
```

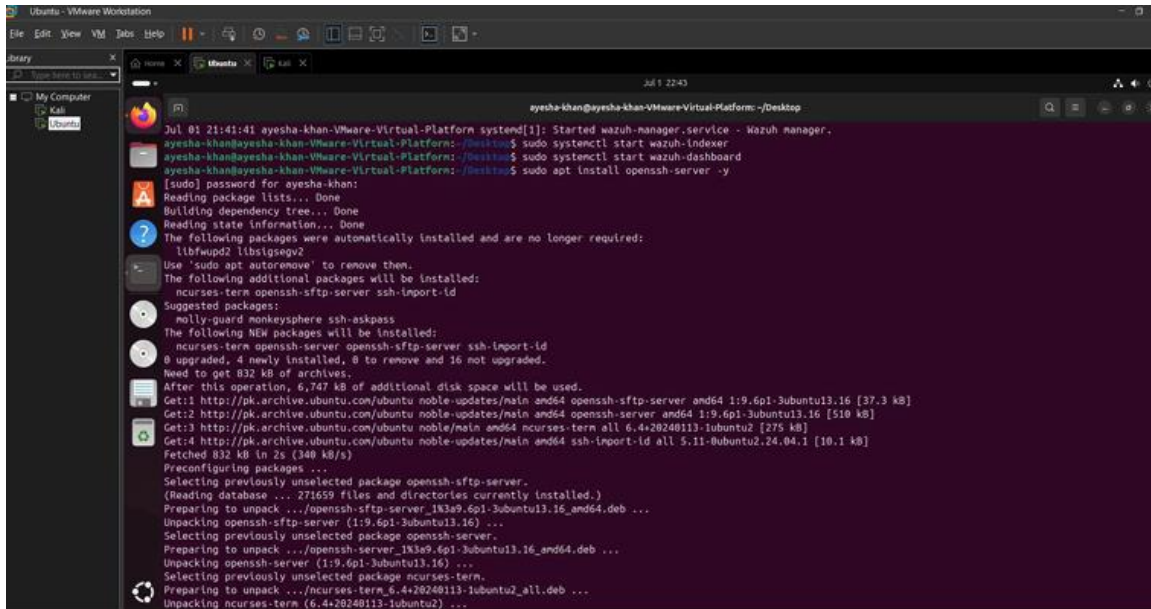
```
Jun 29 20:11:33 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstrap.Open
Jun 29 20:11:33 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap
Jun 29 20:11:33 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: System::setSecurityManager will be removed in a future release
Jun 29 20:11:36 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: Jun 29, 2026 8:11:36 PM sun.util.locale.provider.LocaleProviderAdapter <clinit>
Jun 29 20:11:36 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: COMPAT locale provider will be removed in a future release
Jun 29 20:11:37 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: A terminally deprecated method in java.lang.System has been called
Jun 29 20:11:37 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstrap.Secur
Jun 29 20:11:37 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap
Jun 29 20:11:37 ayesha-khan-Virtual-Platform systemd-entrypoint[7597]: WARNING: System::setSecurityManager will be removed in a future release
Jun 29 20:12:13 ayesha-khan-Virtual-Platform systemd[1]: Started wazuh-indexer.service - wazuh-indexer.
[lines 1-21/21 (END)]
[2]# Stopped
ayesha-khan@ayesha-khan-Virtual-Platform:~/Desktop$ sudo systemctl status wazuh-indexer
● wazuh-indexer.service - wazuh-indexer
   Loaded: loaded (/etc/systemd/system/wazuh-indexer.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-29 20:12:41 PKT; 9min ago
     Docs: https://documentation.wazuh.com
    Main PID: 62802 (node)
      Tasks: 11 (limit: 4541)
     Memory: 159.4M (peak: 318.3M swap: 3.0M swap peak: 3.0M)
        CPU: 35.616s
    CGroup: /system.slice/wazuh-indexer.service
           └─62802 /usr/share/wazuh-indexer/node/bin/node --no-warnings --max-http-header-size=65536 --unhandled-rejections=warn /usr/share/wazuh-indexer/sr

Jun 29 20:27:49 ayesha-khan-Virtual-Platform opensearch-dashboards[62802]: {"type":"log","@timestamp":"2026-06-29T15:27:49Z","tags":["info","plugins","wazu
Jun 29 20:27:49 ayesha-khan-Virtual-Platform opensearch-dashboards[62802]: {"type":"log","@timestamp":"2026-06-29T15:27:49Z","tags":["listening","info"],"p
Jun 29 20:27:50 ayesha-khan-Virtual-Platform opensearch-dashboards[62802]: {"type":"log","@timestamp":"2026-06-29T15:27:50Z","tags":["info","http","server"]
Jun 29 20:27:50 ayesha-khan-Virtual-Platform opensearch-dashboards[62802]: [agentkeepalive:deprecated] options.freeSocketKeepAliveTimeout is deprecated, pl
Jun 29 20:27:52 ayesha-khan-Virtual-Platform opensearch-dashboards[62802]: {"type":"log","@timestamp":"2026-06-29T15:27:52Z","tags":["info","plugins","wazu
Jun 29 20:27:52 ayesha-khan-Virtual-Platform opensearch-dashboards[62802]: {"type":"response","@timestamp":"2026-06-29T15:27:50Z","tags":[""],"pid":62802,"me
Jun 29 20:30:01 ayesha-khan-Virtual-Platform opensearch-dashboards[62802]: {"type":"log","@timestamp":"2026-06-29T15:30:01Z","tags":["error","opensearch"],"
Jun 29 20:30:01 ayesha-khan-Virtual-Platform opensearch-dashboards[62802]: {"type":"log","@timestamp":"2026-06-29T15:30:01Z","tags":["info","plugins","wazu
Jun 29 20:30:01 ayesha-khan-Virtual-Platform opensearch-dashboards[62802]: {"type":"log","@timestamp":"2026-06-29T15:30:01Z","tags":["info","plugins","wazu
Jun 29 20:33:28 ayesha-khan-Virtual-Platform opensearch-dashboards[62802]: {"type":"error","@timestamp":"2026-06-29T15:33:28Z","tags":["connection","client"]
[lines 1-20/20 (END)]
```

VERIFICATION OF WAZUH SERVICES:

After completing the installation, it is important to verify that all Wazuh services are running correctly. This is done using the systemctl status command in Ubuntu. The following services should be checked:

- wazuh-manager
- wazuh-indexer
- wazuh-dashboard



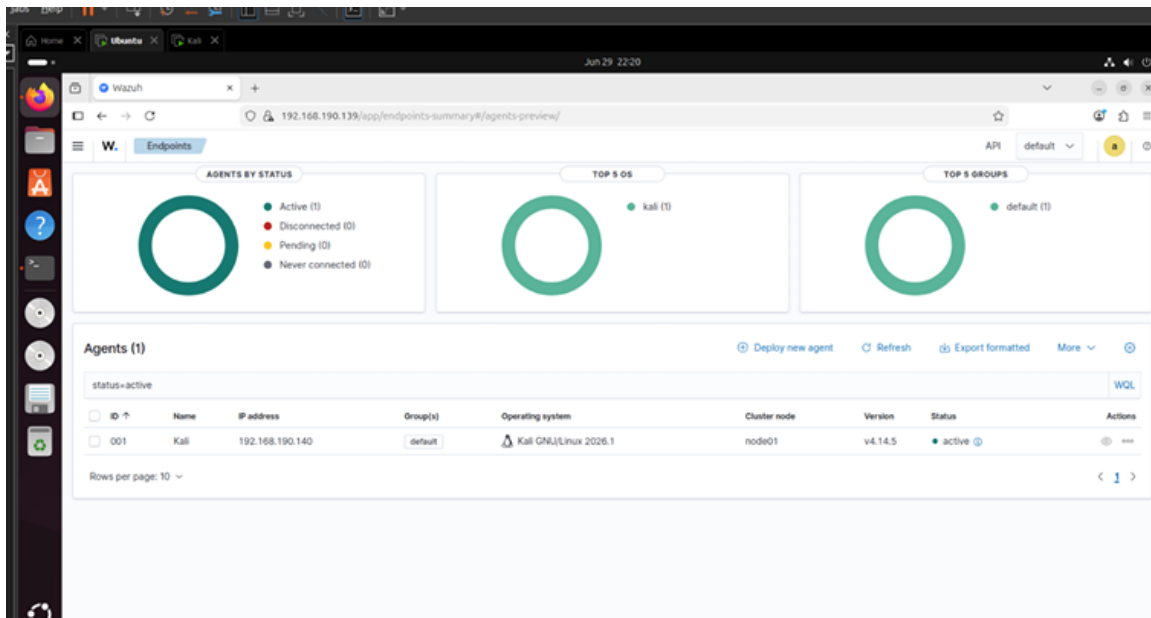
```
Jul 01 21:41:41 ayesha-khan-VMware-Virtual-Platform systemd[1]: Started wazuh-manager.service - Wazuh manager.
ayesha-khan@ayesha-khan-VMware-Virtual-Platform:~/Desktop$ sudo systemctl start wazuh-indexer
ayesha-khan@ayesha-khan-VMware-Virtual-Platform:~/Desktop$ sudo systemctl start wazuh-dashboard
ayesha-khan@ayesha-khan-VMware-Virtual-Platform:~/Desktop$ sudo apt install openssh-server -y
[sudo] password for ayesha-khan:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwpd2 libsigsegv2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 16 not upgraded.
Need to get 832 kB of archives.
After this operation, 6,747 kB of additional disk space will be used.
Get:1 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-server amd64 1:9.6p1-3ubuntu13.16 [37.3 kB]
Get:2 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-server amd64 1:9.6p1-3ubuntu13.16 [510 kB]
Get:3 http://pk.archive.ubuntu.com/ubuntu noble/main amd64 ncurses-term all 6.4+20240113-1ubuntu2 [275 kB]
Get:4 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 ssh-import-id all 5.11-0ubuntu2.24.04.1 [10.1 kB]
Fetched 832 kB in 2s (340 kB/s)
Preconfiguring packages ...
Selecting previously unselected package openssh-sftp-server.
(Reading database ... 271659 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1:9.6p1-3ubuntu13.16_amd64.deb ...
Unpacking openssh-sftp-server (1:9.6p1-3ubuntu13.16) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1:9.6p1-3ubuntu13.16_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.16) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu2_all.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu2) ...
```

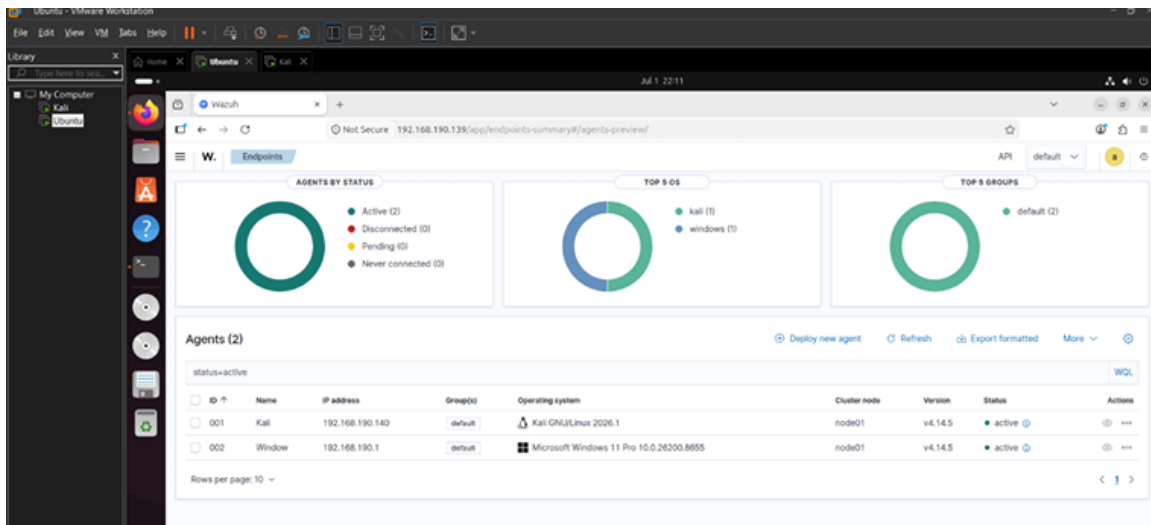
Each service should show the status active (running). This confirms that the SIEM components are functioning correctly and are ready to receive and process security data.

ACCESSING THE WAZUH DASHBOARD:

Once the services are running, the Wazuh Dashboard can be accessed through a web browser using the IP address of the Ubuntu Server. The dashboard is accessed using the following format: The dashboard provides access to several sections including:

- Overview
- Agents
- Security Events
- Threat Detection
- System Monitoring





Wazuh Agent Deployment

Introduction

A **Wazuh Agent** is a small software installed on a computer or server to collect security information. It monitors system activities, gathers logs, and sends them to the **Wazuh Manager** for analysis. Each device that needs monitoring must have a Wazuh agent installed. This helps detect security threats and monitor the health of the system.

Functions of a Wazuh Agent

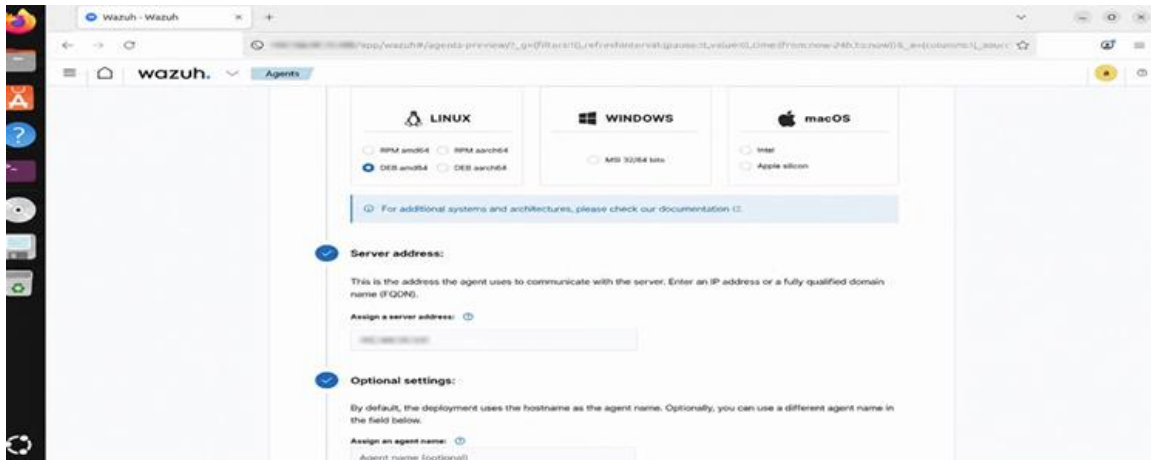
- Collects system and application logs
- Monitors security events
- Detects suspicious activities
- Performs File Integrity Monitoring (FIM)
- Sends data securely to the Wazuh Manager

Agent Deployment Using the Wazuh Dashboard

The **Wazuh Dashboard** provides a simple deployment wizard to install agents. It allows users to select the operating system, enter the Wazuh Manager's IP address, and generate the installation command automatically. This method makes the installation easier and reduces configuration errors.

Deploying the Agent on Kali Linux

Kali Linux is used as a testing machine in the virtual lab. The Wazuh agent was installed using the deployment command generated from the Wazuh Dashboard. After the installation, the agent successfully connected to the Wazuh Manager and started sending security logs for monitoring.



When the Wazuh agent is deployed on Kali Linux, it begins monitoring the system for security-related events. The agent collects logs from the operating system and forwards them to the Wazuh Manager for analysis. Within a short period of time, the agent should appear in the **Wazuh dashboard with an Active status**.

```
shariff@kali:~$ sudo apt install wazuh-agent
Hit:1 http://kali.kali.org/kali kali-rolling InRelease
Get:2 http://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb
Fetched 5,128 kB in 2s (4,235 kB/s)
Selecting previously unselected package wazuh-agent.
(Reading database ... 53920 files and directories currently installed.)
Preparing to unpack wazuh-agent_4.7.5-1_amd64.deb ...
Unpacking wazuh-agent (4.7.5-1) ...
Setting up wazuh-agent (4.7.5-1) ...

shariff@kali:~$ sudo systemctl daemon-reload

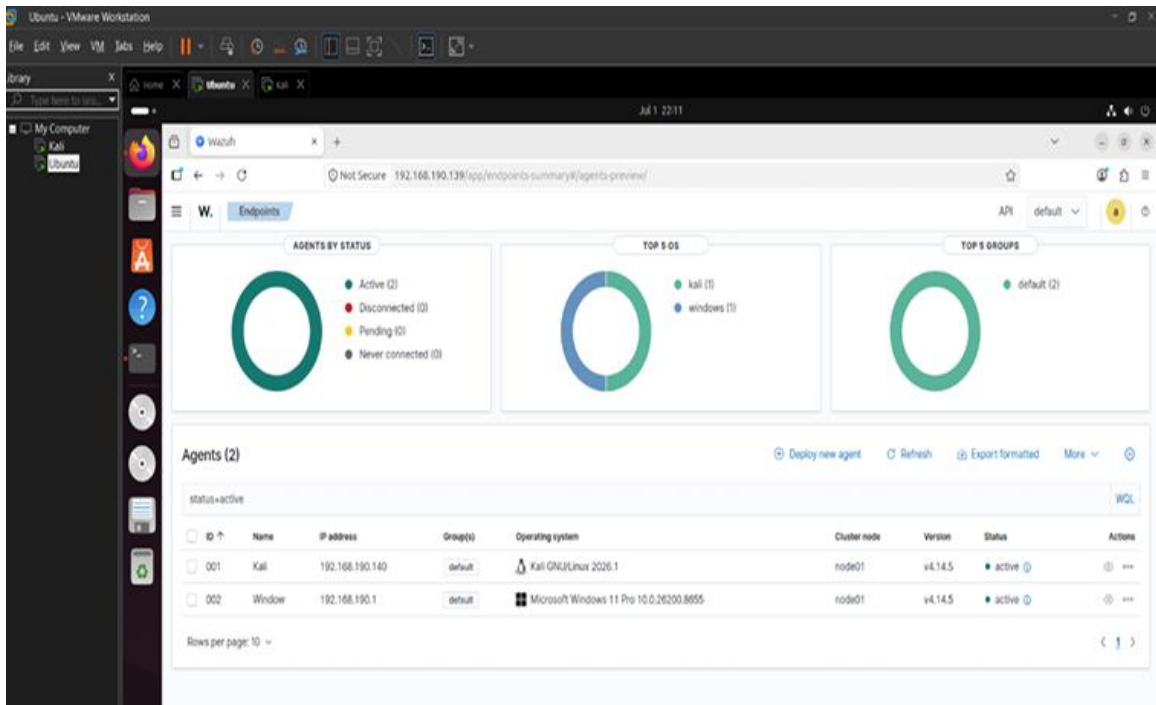
shariff@kali:~$ sudo touch /etc/default/wazuh-agent
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /usr/lib/systemd/system/wazuh-agent.service.

shariff@kali:~$ sudo systemctl start wazuh-agent

shariff@kali:~$ sudo systemctl status wazuh-agent
wazuh-agent.service - wazuh-agent
Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: disabled)
Active: active (running) since Mon 2023-06-23 16:09:05 PDT; 3min 42s ago
Invocation: 4219cd0f080b26f00e0e00fda023
Process: 11182 ExecStartPre=/usr/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
Main PID: 11182 (wazuh-agent)
Tasks: 14 (limit: 9409)
Memory: 51.9M
CGroup: /system.slice/wazuh-agent.service
```

After installation, the **agent service must be started** to activate monitoring. Once the service is running, the agent attempts to establish a secure connection with the manager. This status confirms that:

- The agent is successfully connected to the manager
- Data transmission is working correctly
- Monitoring is active



DEPLOYING THE AGENT ON WINDOW 11:

Windows 11 is one of the most widely used operating systems in enterprise environments. Monitoring Windows endpoints is essential because they often contain sensitive user and organizational data.

The screenshot shows the 'Deploy new agent' form in the Wazuh dashboard. The form is divided into three sections: 'Select the package to download and install on your system:', 'Server address:', and 'Optional settings:'. The 'Select the package' section has three tabs: LINUX, WINDOWS, and macOS. The 'WINDOWS' tab is selected, showing the 'MSI 32-bit x64' package. The 'Server address' section has a text input field with '192.168.190.138' and a 'Remember server address' checkbox. The 'Optional settings' section has a text input field for 'Assign an agent name' with 'Window' entered.

Deploy new agent

Select the package to download and install on your system:

- LINUX**
 - ☐ RPM x86_64
 - ☐ RPM x86_64
 - ☐ DEB amd64
 - ☐ DEB x86_64
- WINDOWS**
 - ☒ MSI 32-bit x64
- macOS**
 - ☐ Intel
 - ☐ Apple silicon

For additional systems and architectures, please check our [documentation](#).

Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

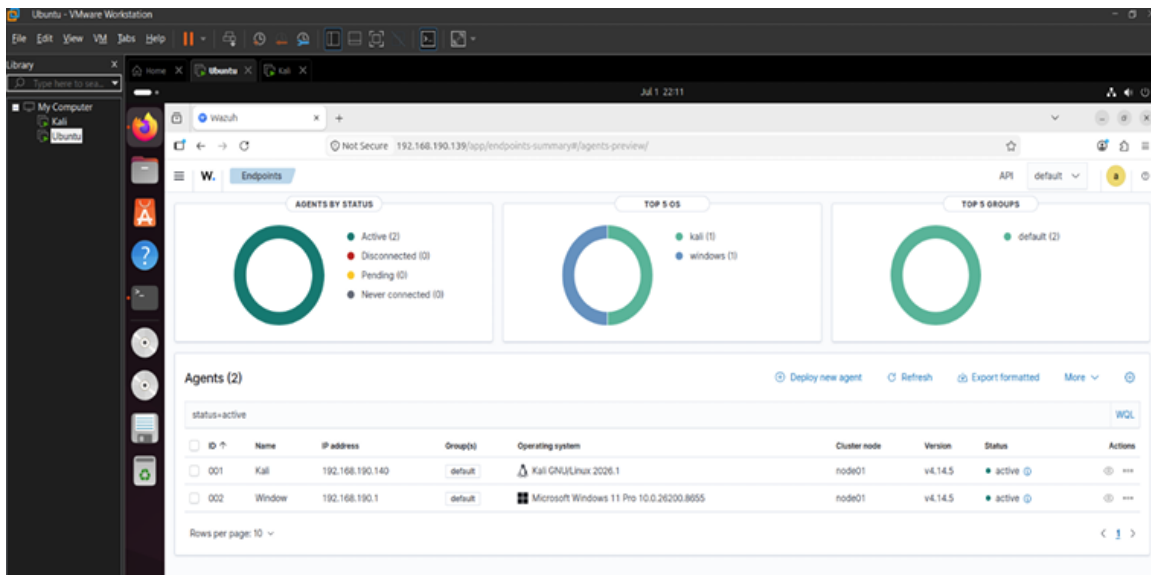
Assign a server address:

☐ Remember server address

Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name:



- The Wazuh deployment wizard generates a PowerShell installation command for Windows systems.

GENERATE SECURITY ALERT AND VERIFY ALERT:

SSH Authentication Failure: An SSH authentication failure occurs when a user attempts to log in to a remote system using the SSH protocol, but the authentication process fails due to invalid credentials, an incorrect password, or a non-existent user account.

- **Event Type:** SSH Authentication Failure
- **Source IP:** 192.168.190.140 (Kali Agent)
- **Target Server:** Ubuntu Wazuh Manager
- **Username Attempted:** Iqra
- **Decoder:** sshd
- **Rule ID:** 5710
- **Severity Level:** 5
- **Description:** Attempt to login using a non-existent user

• MITRE Techniques:

- o T1110.001 – Password Guessing
- o T1021.004 – SSH

Generated Event:

Failed password for invalid user iqra from 192.168.190.140 port 55426 ssh2.

Verification: A failed login attempt using the username iqra was generated from the Kali machine. The Wazuh Agent collected the security event and forwarded it to the Wazuh Manager. The Wazuh Manager analyzed the event and generated a security alert. The alert was successfully verified on the Wazuh Dashboard.

```
Kali - VMware Workstation
File Edit View VM Tabs Help
Library
Total time to search:
My Computer
Ubuntu
Apps Places
Jul 1 1:36 PM
iqrakali:~
$ sudo systemctl status wazuh-agent
[sudo] password for iqrakali:
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-07-01 11:59:36 EDT; 1h 22min ago
     Invocation: e55060320b4d99777c559392e4ef
     Process: 3258 ExecStartPre=/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 32 (limit: 19386)
   Memory: 104.0M (peak: 114.9M)
      CPU: 7min 9.60s
   CGroup: /system.slice/wazuh-agent.service
           └─9935 /var/ossec/bin/wazuh-execd
             └─9947 /var/ossec/bin/wazuh-execd
               └─9963 /var/ossec/bin/wazuh-syscheckd
                 └─10043 /var/ossec/bin/wazuh-logcollector
                   └─10057 /var/ossec/bin/wazuh-modulesd

Jul 01 11:59:39 kali env[925]: Deleting PID file "/var/ossec/var/run/wazuh-syscheckd-1199.pid" not used...
Jul 01 11:59:39 kali env[925]: Deleting PID file "/var/ossec/var/run/wazuh-agent-1087.pid" not used...
Jul 01 11:59:39 kali env[925]: Deleting PID file "/var/ossec/var/run/wazuh-execd-1003.pid" not used...
Jul 01 11:59:39 kali env[925]: Started wazuh-modulesd...
Jul 01 11:59:39 kali env[925]: Started wazuh-execd...
Jul 01 11:59:39 kali env[925]: Started wazuh-syscheckd...
Jul 01 11:59:39 kali env[925]: Started wazuh-logcollector...
Jul 01 11:59:39 kali env[925]: Started wazuh-modulesd...
Jul 01 11:59:39 kali system[1]: Completed.
Jul 01 11:59:36 kali systemd[1]: Started wazuh-agent.service - Wazuh agent.

[iqrakali:~]
$ ping 192.168.190.139
PING 192.168.190.139 (192.168.190.139) 56(84) bytes of data:
From 192.168.190.139: icmp_seq=1 ttl=128 time=0.923 ms
...
192.168.190.139 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/ndev = 0.923/0.923/0.923/0.000 ms

[iqrakali:~]
$ sudo systemctl restart wazuh-agent
[iqrakali:~]
$ sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-01 13:36:44 EDT; 3s ago

[iqrakali:~]
```

Wazuh Threat Hunting

632 hits

Jun 30, 2026 @ 22:21:20.863 - Jul 1, 2026 @ 22:21:20.864

Export Formatted | Reset view | 682 available fields | Columns | Density | 1 fields sorted | Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Jul 1, 2026 @ 22:29:28.019	iqrakali-batool-VMware-Virtual-Platform	syslog: User missed the password more than one time	10	2502
Jul 1, 2026 @ 22:29:25.019	iqrakali-batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existent user	5	5710
Jul 1, 2026 @ 22:29:21.017	iqrakali-batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existent user	5	5710
Jul 1, 2026 @ 22:29:11.010	iqrakali-batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existent user	5	5710
Jul 1, 2026 @ 22:29:09.011	iqrakali-batool-VMware-Virtual-Platform	PAM: User login failed.	5	5503
Jul 1, 2026 @ 22:29:05.008	iqrakali-batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existent user	5	5710
Jul 1, 2026 @ 22:27:06.988	iqrakali-batool-VMware-Virtual-Platform	PAM: Login session closed.	3	5502
Jul 1, 2026 @ 22:27:04.956	iqrakali-batool-VMware-Virtual-Platform	PAM: Login session opened.	3	5501
Jul 1, 2026 @ 22:27:04.955	iqrakali-batool-VMware-Virtual-Platform	Successful sudo to ROOT executed.	3	5402
Jul 1, 2026 @ 22:26:28.944	iqrakali-batool-VMware-Virtual-Platform	PAM: Login session opened.	3	5501
Jul 1, 2026 @ 22:26:28.944	iqrakali-batool-VMware-Virtual-Platform	PAM: Login session closed.	3	5502
Jul 1, 2026 @ 22:26:20.427	Window	Software protection service scheduled successfully.	3	60642
Jul 1, 2026 @ 22:26:20.406	iqrakali-batool-VMware-Virtual-Platform	Dpkg (Debian Package) half configured.	7	2904
Jul 1, 2026 @ 22:26:20.406	iqrakali-batool-VMware-Virtual-Platform	New dpkg (Debian Package) installed.	7	2902
Jul 1, 2026 @ 22:26:20.402	iqrakali-batool-VMware-Virtual-Platform	New dpkg (Debian Package) installed.	7	2902

Rows per page: 15

Wazuh Threat Hunting interface showing a list of events and document details.

Table View:

timestamp	agent.name	rule.description
Jul 1, 2026 @ 22:29:28.019	iqra batool-VMware-Virtual-Platform	syslog: User missed the password more than one time
Jul 1, 2026 @ 22:29:25.019	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existing user
Jul 1, 2026 @ 22:29:21.017	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existing user
Jul 1, 2026 @ 22:29:11.010	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existing user
Jul 1, 2026 @ 22:29:09.011	iqra batool-VMware-Virtual-Platform	PAM: User login failed.
Jul 1, 2026 @ 22:29:05.008	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existing user
Jul 1, 2026 @ 22:27:06.988	iqra batool-VMware-Virtual-Platform	PAM: Login session closed.
Jul 1, 2026 @ 22:27:04.956	iqra batool-VMware-Virtual-Platform	PAM: Login session opened.
Jul 1, 2026 @ 22:27:04.955	iqra batool-VMware-Virtual-Platform	Successful sudo to ROOT executed.
Jul 1, 2026 @ 22:26:28.944	iqra batool-VMware-Virtual-Platform	PAM: Login session opened.
Jul 1, 2026 @ 22:26:28.944	iqra batool-VMware-Virtual-Platform	PAM: Login session closed.
Jul 1, 2026 @ 22:26:20.427	Windows	Software protection service scheduled
Jul 1, 2026 @ 22:26:20.406	iqra batool-VMware-Virtual-Platform	Dpkg (Debian Package) half configured
Jul 1, 2026 @ 22:26:20.406	iqra batool-VMware-Virtual-Platform	New dpkg (Debian Package) installed.
Jul 1, 2026 @ 22:26:20.402	iqra batool-VMware-Virtual-Platform	New dpkg (Debian Package) installed.

Rows per page: 15

Document Details:

View surrounding documents | View single document

Table | JSON

Field	Value
index	wazuh-alerts-4.x-2026_07_01
agent.id	000
agent.name	iqra batool-VMware-Virtual-Platform
decoder.name	sshd
full_log	Jul 01 17:29:28 iqra batool-VMware-Virtual-Platform sshd[14378]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.190.140
id	1762624490.2236312
input.type	log
location	journal0
manager.name	iqra batool-VMware-Virtual-Platform
predecoder.hostname	iqra batool-VMware-Virtual-Platform
predecoder.program_name	sshd
predecoder.timestamp	Jul 01 17:29:28
rule.description	syslog: User missed the password more than one time
rule.firetimes	1
rule.gdpr	IV.35.7.d, IV.32.2
rule.gp13	7.1
rule.groups	syslog, access_control, authentication_failed

Wazuh Threat Hunting interface showing a list of events and document details.

Table View:

timestamp	agent.name	rule.description
Jul 1, 2026 @ 22:29:28.019	iqra batool-VMware-Virtual-Platform	syslog: User missed the password more than one time
Jul 1, 2026 @ 22:29:25.019	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existing user
Jul 1, 2026 @ 22:29:21.017	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existing user
Jul 1, 2026 @ 22:29:11.010	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existing user
Jul 1, 2026 @ 22:29:09.011	iqra batool-VMware-Virtual-Platform	PAM: User login failed.
Jul 1, 2026 @ 22:29:05.008	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-existing user
Jul 1, 2026 @ 22:27:06.988	iqra batool-VMware-Virtual-Platform	PAM: Login session closed.
Jul 1, 2026 @ 22:27:04.956	iqra batool-VMware-Virtual-Platform	PAM: Login session opened.
Jul 1, 2026 @ 22:27:04.955	iqra batool-VMware-Virtual-Platform	Successful sudo to ROOT executed.
Jul 1, 2026 @ 22:26:28.944	Window	PAM: Login session opened.
Jul 1, 2026 @ 22:26:28.427	iqra batool-VMware-Virtual-Platform	PAM: Login session closed.
Jul 1, 2026 @ 22:26:20.408	iqra batool-VMware-Virtual-Platform	Software protection service scheduled
Jul 1, 2026 @ 22:26:20.406	iqra batool-VMware-Virtual-Platform	Dpkg (Debian Package) half configured
Jul 1, 2026 @ 22:26:20.406	iqra batool-VMware-Virtual-Platform	New dpkg (Debian Package) installed.
Jul 1, 2026 @ 22:26:20.402	iqra batool-VMware-Virtual-Platform	New dpkg (Debian Package) installed.

Rows per page: 15

Document Details:

View surrounding documents | View single document

Table | JSON

Field	Value
manager.name	iqra batool-VMware-Virtual-Platform
predecoder.hostname	iqra batool-VMware-Virtual-Platform
predecoder.program_name	sshd
predecoder.timestamp	Jul 01 17:29:09
rule.description	PAM: User login failed.
rule.firetimes	1
rule.gdpr	IV.35.7.d, IV.32.2
rule.gp13	7.0
rule.groups	pwd, syslog, authentication_failed
rule.hipaa	164.312.b
rule.id	5583
rule.level	5
rule.mail	false
rule.mitre.id	T1110.005
rule.mitre.tactic	Credential Access
rule.mitre.technique	Password Guessing
rule.nist_800_53	AU.14, AC.7
rule.pci_dss	10.2.4, 10.2.5
rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3
timestamp	Jul 1, 2026 @ 22:29:09.011

The screenshot displays the Wazuh Threat Hunting interface. On the left, a table lists alerts with columns for timestamp, agent name, and rule description. The right pane shows the 'Document Details' for a selected alert, including fields like _index, agent.id, agent.name, data.srcip, data.srcuser, decoder.name, decoder.parent, full_log, id, input.type, location, manager.name, predecoder.hostname, predecoder.program_name, predecoder.timestamp, rule.description, rule.firedtimes, and rule.gdpr.

timestamp	agent.name	rule.description
Jul 1, 2026 @ 22:29:28.019	iqra batool-VMware-Virtual-Platform	syslog: User missed the password mo
Jul 1, 2026 @ 22:29:25.019	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-exi
Jul 1, 2026 @ 22:29:21.017	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-exi
Jul 1, 2026 @ 22:29:11.010	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-exi
Jul 1, 2026 @ 22:29:09.011	iqra batool-VMware-Virtual-Platform	PAM: User login failed.
Jul 1, 2026 @ 22:29:05.008	iqra batool-VMware-Virtual-Platform	sshd: Attempt to login using a non-exi
Jul 1, 2026 @ 22:27:06.988	iqra batool-VMware-Virtual-Platform	PAM: Login session closed.
Jul 1, 2026 @ 22:27:04.956	iqra batool-VMware-Virtual-Platform	PAM: Login session opened.
Jul 1, 2026 @ 22:27:04.955	iqra batool-VMware-Virtual-Platform	Successful sudo to ROOT executed.
Jul 1, 2026 @ 22:26:28.944	Window	PAM: Login session opened.
Jul 1, 2026 @ 22:26:28.427	iqra batool-VMware-Virtual-Platform	PAM: Login session closed.
Jul 1, 2026 @ 22:26:20.403	iqra batool-VMware-Virtual-Platform	Software protection service schedul
Jul 1, 2026 @ 22:26:20.406	iqra batool-VMware-Virtual-Platform	Dpkg (Debian Package) half configu
Jul 1, 2026 @ 22:26:20.406	iqra batool-VMware-Virtual-Platform	New dpkg (Debian Package) installed.
Jul 1, 2026 @ 22:26:20.402	iqra batool-VMware-Virtual-Platform	New dpkg (Debian Package) installed.

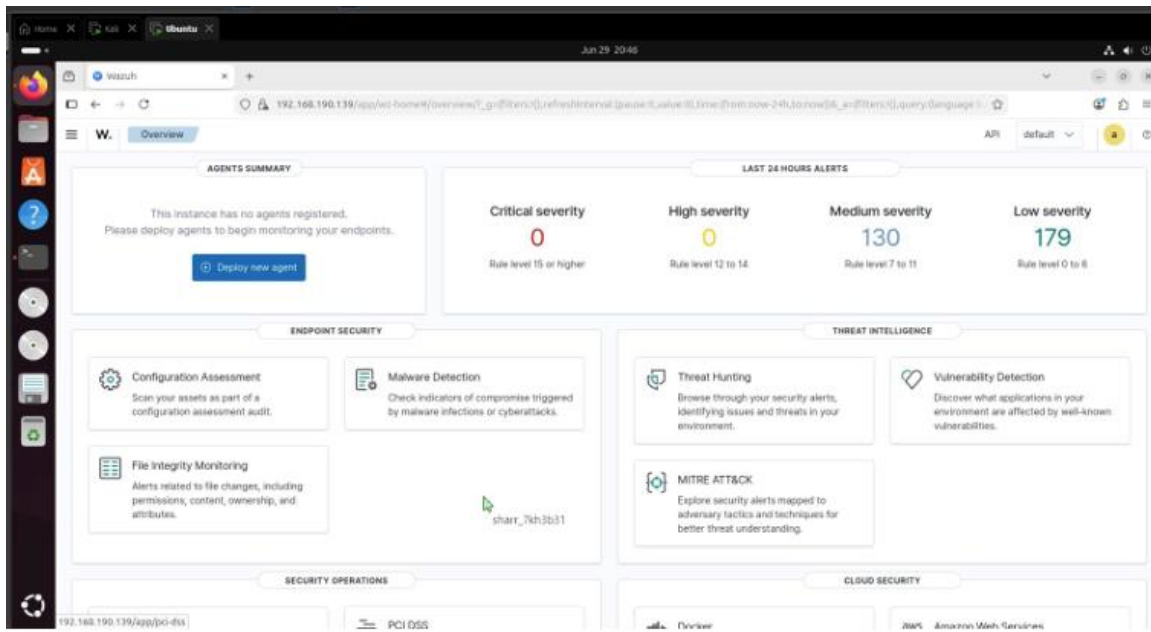
Field	Value
_index	wazuh-alerts-4.x-2026.07.01
agent.id	000
agent.name	iqra batool-VMware-Virtual-Platform
data.srcip	192.168.190.140
data.srcuser	iqra
decoder.name	sshd
decoder.parent	sshd
full_log	Jul 01 17:20:19 iqra batool-VMware-Virtual-Platform sshd[14378]: Failed password for invalid user iqra from 192.168.190.140 port 54340 ssh2
id	1762020410.2233152
input.type	log
location	journal0
manager.name	iqra batool-VMware-Virtual-Platform
predecoder.hostname	iqra batool-VMware-Virtual-Platform
predecoder.program_name	sshd
predecoder.timestamp	Jul 01 17:20:19
rule.description	sshd: Attempt to login using a non-existent user
rule.firedtimes	3
rule.gdpr	IV_35.7.d, IV_32.2

CONFIGURE AND VALIDATE FILE INTEGRITY MONITORING (FIM)

File Integrity Monitoring (FIM) is a security technique used to detect changes made to important files and directories in a system. In a modern security environment, attackers frequently modify system files, configuration files, or logs to maintain persistence, hide malicious activity, or escalate privileges.

FIM continuously monitors selected files and directories and generates alerts whenever a file is:

- Created
- Modified
- Deleted



The primary purpose of FIM is to provide visibility into unauthorized or suspicious file changes. This allows security analysts to quickly detect possible system compromise.

In the Wazuh platform, the component responsible for File Integrity Monitoring is called syscheck. The syscheck module monitors configured paths and generates alerts whenever changes occur.

Accessing the ossec.conf Configuration File

The configuration of File Integrity Monitoring in Wazuh is performed inside the **ossec.conf** configuration file.

This file contains the core configuration settings for the Wazuh agent, including:

- Log monitoring
- Rootkit detection
- File integrity monitoring
- System checks

Inside this configuration file, the syscheck block controls how file monitoring works.

Editing the ossec.conf file allows the administrator to:

- Define which directories should be monitored
- Enable real-time monitoring
- Control how frequently file scans occur
- Configure whether detailed file changes should be recorded

Proper configuration of this file ensures that the Wazuh agent monitors critical parts of the operating system.

Selecting Critical Directories for Monitoring

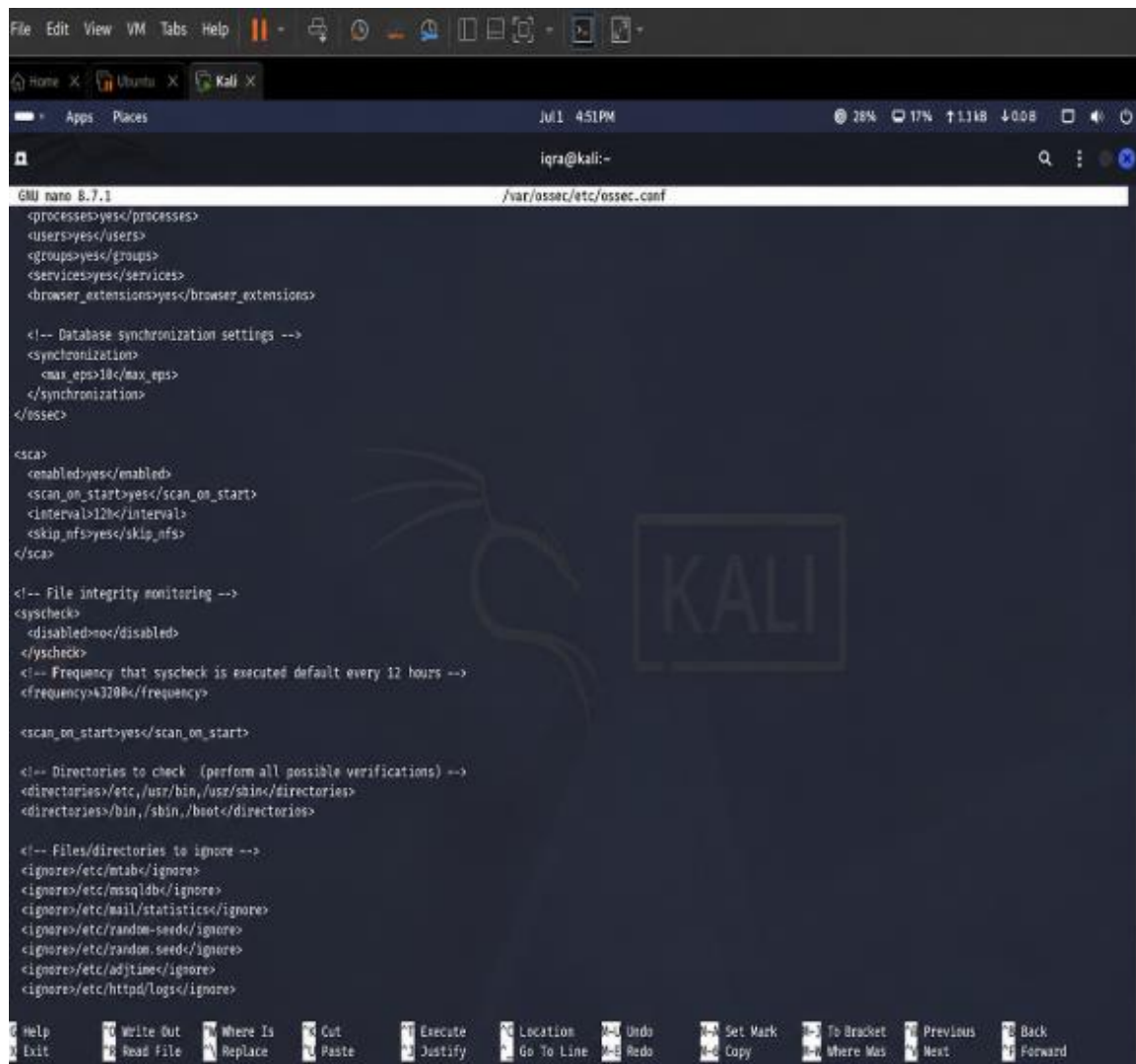
For effective security monitoring, it is important to track directories that contain sensitive files.

In this task, the following directories were configured

LINUX Directories

The /etc directory contains critical system configuration files such as:

- user account settings
- network configuration
- service configuration
- security policies



```
GNU nano 8.7.1 /var/ossec/etc/ossec.conf
<processes>yes</processes>
<users>yes</users>
<groups>yes</groups>
<services>yes</services>
<browser_extensions>yes</browser_extensions>

<!-- Database synchronization settings -->
<synchronization>
  <max_eps>10</max_eps>
</synchronization>
</ossec>

<sca>
  <enabled>yes</enabled>
  <scan_on_start>yes</scan_on_start>
  <interval>120</interval>
  <skip_nfs>yes</skip_nfs>
</sca>

<!-- File integrity monitoring -->
<syscheck>
  <disabled>no</disabled>
</syscheck>
<!-- Frequency that syscheck is executed default every 12 hours -->
<frequency>43200</frequency>

<scan_on_start>yes</scan_on_start>

<!-- Directories to check (perform all possible verifications) -->
<directories>/etc,/usr/bin,/usr/sbin</directories>
<directories>/bin,/sbin,/boot</directories>

<!-- Files/directories to ignore -->
<ignore>/etc/mtab</ignore>
<ignore>/etc/mssqldb</ignore>
<ignore>/etc/mail/statistics</ignore>
<ignore>/etc/random-seed</ignore>
<ignore>/etc/random-seed</ignore>
<ignore>/etc/adjtime</ignore>
<ignore>/etc/httpd/logs</ignore>
```



```

(iqra@kali)-[~]
$ sudo nano /var/ossec/etc/ossec.conf

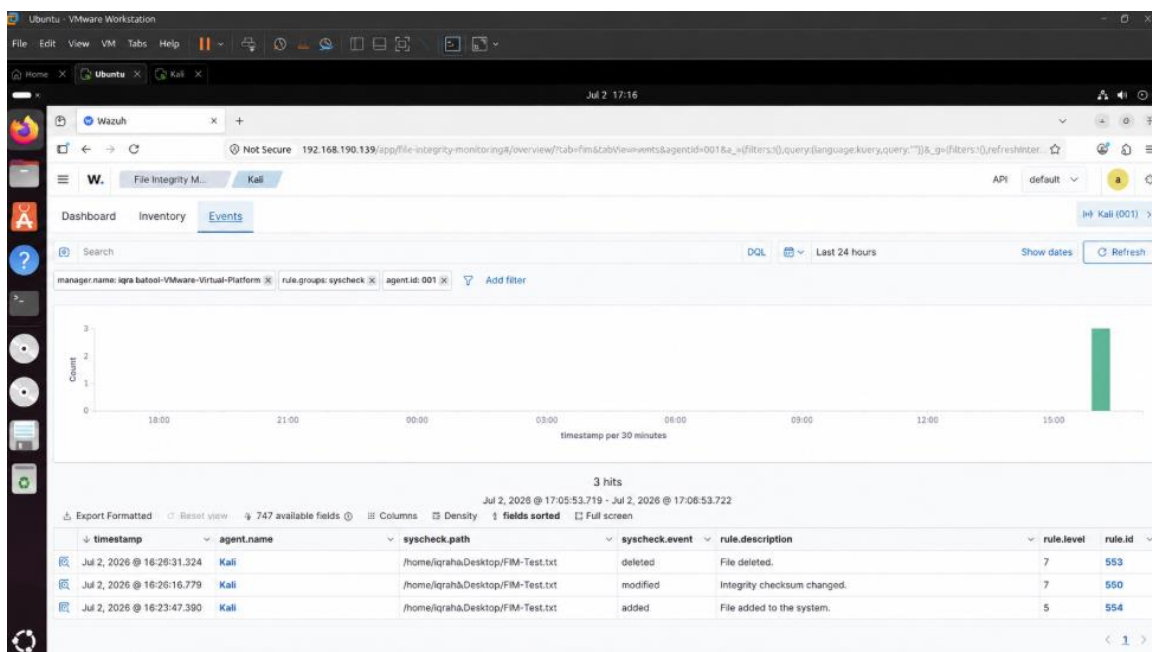
(iqra@kali)-[~]
$ sudo systemctl restart wazuh-agent

(iqra@kali)-[~]
$ sudo systemctl status wazuh-agent
wazuh-agent.service - Wazuh agent
Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: disabled)
Active: active (running) since Thu 2026-07-02 07:05:46 EDT; 48s ago
Invocation: 2d6981bdd34b4a98a71b7134cbcf067
Process: 100600 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
Tasks: 30 (limit: 9924)
Memory: 21.1M (peak: 28.6M)
CPU: 52.727s
CGroup: /system.slice/wazuh-agent.service
├─100622 /var/ossec/bin/wazuh-execd
├─100645 /var/ossec/bin/wazuh-agentd
├─100665 /var/ossec/bin/wazuh-syscheckd
├─100685 /var/ossec/bin/wazuh-logcollector
├─100708 /var/ossec/bin/wazuh-modulesd
├─102413 sh -c -- "/bin/ps -p 4485 2> /dev/null"
└─102414 /bin/ps -p 4485

Jul 02 07:05:38 kali systemd[1]: Starting wazuh-agent.service - Wazuh agent...
Jul 02 07:05:38 kali env[100600]: Starting Wazuh v4.14.5...
Jul 02 07:05:39 kali env[100600]: Started wazuh-execd...
Jul 02 07:05:40 kali env[100600]: Started wazuh-agentd...
Jul 02 07:05:41 kali env[100600]: Started wazuh-syscheckd...
Jul 02 07:05:43 kali env[100600]: Started wazuh-logcollector...
Jul 02 07:05:44 kali env[100600]: Started wazuh-modulesd...
Jul 02 07:05:46 kali env[100600]: Completed.
Jul 02 07:05:46 kali systemd[1]: Started wazuh-agent.service - Wazuh agent.

(iqra@kali)-[~]
$

```



WINDOWS DIRECTORY :

The System32 directory is one of the most critical directories in Windows systems. It contains:

- core system
- files operating system binaries
- DLL libraries
- essential system utilities

```
File Edit View
<!-- File integrity monitoring -->
<syscheck>

<disabled>yes</disabled>
<scan_on_start>yes</scan_on_start>
<interval>12h</interval>
<skip_nfs>yes</skip_nfs>
</syscheck>

<!-- Frequency that syscheck is executed default every 12 hours -->
<frequency>600</frequency>

<!-- Default files to be monitored. -->
<directories recursion_level="0" restrict="regedit.exe$system.ini$win.ini$%WINDIR%/directories>

<directories recursion_level="0" restrict="at.exe$attrib.exe$cacis.exe$cmd.exe$eventcreate.exe$ftp.exe$lsass.exe$net.exe$net1.exe$netsh.exe$reg.exe$regedit32.exe$regsvr32.exe$runas.exe$sc.exe$schtasks.exe$sethc.exe$subst.exe$%WINDIR%/SystemNative/directories>
<directories recursion_level="0" restrict="%WINDIR%/SystemNative/drivers/etc/directories>
<directories recursion_level="0" restrict="wmic.exe$%WINDIR%/SystemNative/wbem/directories>
<directories recursion_level="0" restrict="powershell.exe$%WINDIR%/SystemNative/WindowsPowerShell/v1.0/directories>
<directories recursion_level="0" restrict="winrm.vbs$%WINDIR%/SystemNative/directories>

<!-- 32-bit programs. -->
<directories recursion_level="0" restrict="at.exe$attrib.exe$cacis.exe$cmd.exe$eventcreate.exe$ftp.exe$lsass.exe$net.exe$net1.exe$netsh.exe$reg.exe$regedit32.exe$regsvr32.exe$runas.exe$sc.exe$schtasks.exe$sethc.exe$subst.exe$%WINDIR%/System32/directories>
<directories recursion_level="0" restrict="%WINDIR%/System32/drivers/etc/directories>
<directories recursion_level="0" restrict="wmic.exe$%WINDIR%/System32/wbem/directories>
<directories recursion_level="0" restrict="powershell.exe$%WINDIR%/System32/WindowsPowerShell/v1.0/directories>
<directories recursion_level="0" restrict="winrm.vbs$%WINDIR%/System32/directories>

<directories realtime="yes">%PROGRAMDATA%/Microsoft/Windows/Start Menu/Programs/Startup/directories>

<ignore>%PROGRAMDATA%/Microsoft/Windows/Start Menu/Programs/Startup/desktop.ini</ignore>

<ignore type="sregex">.log$|.htm$|.jpg$|.png$|.chm$|.pnm$|.evtx$</ignore>

<!-- Windows registry entries to monitor. -->
```

```
Administrator: Windows PowerShell
Windows PowerShell 5.1
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> whoami
sharif\sharr_7kh3b31
PS C:\WINDOWS\system32> Restart-Service Wazuh
PS C:\WINDOWS\system32> Get-Service Wazuh

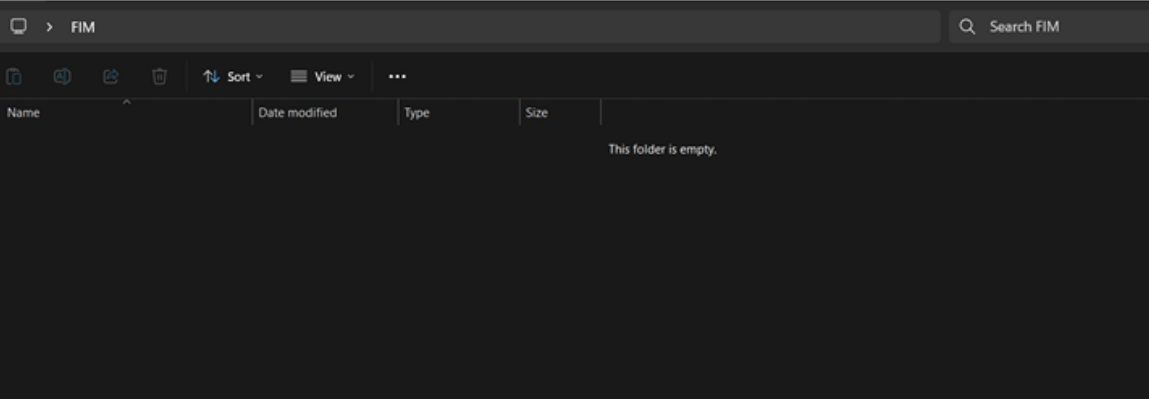
Status Name DisplayName
-----
Running WazuhSvc Wazuh
```

A new file was created inside /etc

The file was modified

The file was deleted

On Windows machine



117 hits
Jul 2, 2026 @ 03:37:53.681 - Jul 3, 2026 @ 03:37:18.681

Export Formatted ☐ Reset view ☐ 744 available fields ☐ Columns ☐ Density ☐ 1 fields sorted ☐ Full screen

timestamp	agent.name	syscheck.path	syscheck.event	rule.description	rule.level	rule.id
Jul 2, 2026 @ 20:15:55.958	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	deleted	Registry Value Entry Deleted.	5	751
Jul 2, 2026 @ 20:15:55.743	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5	750
Jul 2, 2026 @ 20:15:55.743	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Key Integrity Checksum Changed	5	594
Jul 2, 2026 @ 20:15:54.874	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Key Integrity Checksum Changed	5	594
Jul 2, 2026 @ 20:15:52.637	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5	750
Jul 2, 2026 @ 20:15:52.625	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Key Integrity Checksum Changed	5	594
Jul 2, 2026 @ 20:15:52.625	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5	750
Jul 2, 2026 @ 20:15:52.625	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5	750
Jul 2, 2026 @ 20:15:52.615	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Key Integrity Checksum Changed	5	594
Jul 2, 2026 @ 20:15:52.615	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5	750
Jul 2, 2026 @ 20:15:52.600	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5	750
Jul 2, 2026 @ 20:15:52.590	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Key Integrity Checksum Changed	5	594
Jul 2, 2026 @ 20:15:52.590	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Key Integrity Checksum Changed	5	594
Jul 2, 2026 @ 20:15:52.590	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5	750
Jul 2, 2026 @ 20:15:52.403	IGRA	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Ser...	modified	Registry Value Integrity Checksum Changed	5	750

Rows per page: 15

< 1 2 3 4 5 ... 8 >

Document Details

[View surrounding documents](#)[View single document](#)

[Table](#) [JSON](#)

<code>_index</code>	wazuh-alerts-4.x-2026.07.02
<code>agent.id</code>	002
<code>agent.ip</code>	192.168.197.1
<code>agent.name</code>	IQRA
<code>decoder.name</code>	syscheck_registry_value_added
<code>full_log</code>	Registry Value '[x32] HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\TPM\WMI\PcrInfo\0000000275-0000000005\ReplayPcr' added Mode: scheduled
<code>id</code>	1783005349.4313513
<code>input.type</code>	log
<code>location</code>	syscheck
<code>manager.name</code>	iqra-VMware-Virtual-Platform
<code>rule.description</code>	Registry Value Entry Added to the System
<code>rule.firedtimes</code>	11
<code>rule.gdpr</code>	II_5.1.f
<code>rule.gpg13</code>	4.13
<code>rule.groups</code>	ossec, syscheck, syscheck_entry_added, syscheck_registry
<code>rule.hipaa</code>	164.312.c.1, 164.312.c.2
<code>rule.id</code>	752
<code>rule.level</code>	5
<code>rule.mail</code>	false
<code>rule.mitre.id</code>	T1112

Conclusion

This project helped me understand the basic working of a **Security Operations Center (SOC)** using **Wazuh SIEM**. I successfully installed the Wazuh Manager, connected endpoints using the Wazuh Agent, and configured **File Integrity Monitoring (FIM)**. The generated test alerts confirmed that the system was monitoring and detecting security events correctly. Overall, this task improved my practical knowledge of SIEM, endpoint monitoring, and basic security analysis, which are important skills for a SOC or Blue Team intern.

